A COMPETENCY-ASSESSMENT HANDBOOK

Miller's Pyramid for Rasashastra & Bhaishajya Kalpana

Module-by-module competency mapping, assessment design and usage guide for AYPG-RB — Rasashastra and Bhaishajya Kalpana MD/MS Ayurveda, Semesters III to VI

Curriculum	NCISM, Board of Ayurveda — four semesters, four papers, applicable from the 2024–25 batch
Coverage	32 modules · 124 units · 577 session learning objectives · 141 practical training and 163 experiential learning activities
Framework	Miller (1990) four tiers, with the Cruess *Is* tier (2016), mapped onto NCISM Table 3, Column 3F

64	1920	1600	800
CREDITS	NOTIONAL HOURS	FORMATIVE MARKS	UNIVERSITY EXAM

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, assessment design and paper setting.

Contents

PART A — FOUNDATIONS

Chapter 1 Miller's pyramid 5

Chapter 2 The pyramid is already inside your curriculum 6

Chapter 3 Constructive alignment 7

Chapter 4 The assessment toolkit, tier by tier 8

Chapter 5 How to use a module in Part B 9

PART B — SEMESTER 3

Module 1 Chronological Development and Taxonomy of Rasa Dravyas 19 obj · Does 11

Module 2 Puta and Samskara (Transformative processes of drug manufacturing) 21 obj · Does 14

Module 9 Scientific fundamentals of Ayurvedic Pharmaceutical preparations 15 obj · Does 18

Module 10 Principles of pharmaceutical formulation 12 obj · Does 21

Module 17 Guidelines for administration of Rasoushadhi in clinical Practice 22 obj · Does 24

Module 18 Prayogika Bhaishajya Kalpana(Clinical application of Pharmaceutical products) 24 obj · Does 27

Module 25 Artificial intelligence in drug development and Pharmacogenetics; Pharmacogenomics 20 obj · Does 30

Module 26 Pharmacology 16 obj · Does 33

PART B — SEMESTER 4

Module 3 Parada (Mercury) and Its Processing: Ancient wisdom and Scientific advances 17 obj · Does 36

Module 4 Maharasa Dravya, its processing and utility 22 obj · Does 39

Module 11 Mana Paribhasha (Metrology in Ayurvedic Pharmaceutical preparation) 16 obj · Does 42

Module 12 Process validation Protocol 10 obj · Shows how 45

Module 19 SOP and clinical application of Chaturvidha Rasayana 25 obj · Does 48

Module 20 SOP and Clinical application of Ayaskriti, Louha, and Mandura Kalpa 12 obj · Does 51

Module 27 Network pharmacology & Reverse Pharmacology 16 obj · Does 54

Module 28 Pharmaceutical Quality Assurance 19 obj · Does 57

PART B — SEMESTER 5

Module 5 Critical appraisal of Uparasa Dravya and Sadharana Rasa Dravya 24 obj · Does 60

Module 6 Critical appraisal of Dhatu and Updhatu 24 obj · Does 63

Module 13 Krutanna (Ayurvedic Food Preparations) 15 obj · Does 66

Module 14 Ayurvedic Formulations for Holistic Wellbeing, Vitality and Body Care 11 obj · Shows how 69

Module 21 Clinical application of Rasoushadhi according to Srotas and in selected challenging clinical conditions 13 obj · Does 72

Module 22 Yogas from different Kalpanas 27 obj · Does 75

Module 29 Pharmacy Practice 18 obj · Does 79

Module 30 Pharmaceutical Microbiology 18 obj · Does 82

PART B — SEMESTER 6

Module 7 Sudha Varga, Sikata Varga, Visha Varga Dravyas 22 obj · Does 85

Module 8 Appraisal of Ratna and Uparatna and therapeutic significance of Pishti and Druti Kalpana 13 obj · Does 88

Module 15 Disease condition specific selection of formulation 12 obj · Shows how 91

Module 16 Current trends in Ayurvedic Pharmaceutical Industry. 12 obj · Does 94

Module 23 SOP and clinical applications of Churna Kalpana(Powder dosage forms) 26 obj · Does 97

Module 24 Therapeutic significance of Topical dosage forms 11 obj · Shows how 101

Module 31 Regulatory Acts and Rules applicable to Ayurvedic Drugs 31 obj · Does 104

Module 32 Entrepreneurship and Start-Ups 14 obj · Does 108

Every unit is listed with its ceiling in Appendix C2.

PART C — REFERENCE

C1 The four tiers, and what certifies each 112

C2 Ceiling of every module 113

C3 The instruments this curriculum names, by tier 118

C4 Sources, and a note on how this book was made 119

PART A

Foundations

Five short chapters. The first explains what Miller's pyramid is and, just as importantly, what it is not. The second shows that it is already inside your curriculum, in a column most teachers have never had pointed out. The third gives the three rules of constructive alignment, the fourth sets out which instrument can certify which tier, and the fifth explains how to read a unit in Part B. Read straight through once; after that, use the modules.

1. **Miller's pyramid** — what it is, and what it is not
2. **The pyramid is already inside your curriculum** — reading Column 3F
3. **Constructive alignment** — the three rules
4. **The assessment toolkit**, tier by tier
5. **How to use a module in Part B**

↑ Contents

PART A · CHAPTER 1

Miller's pyramid — what it is, and what it is not

In 1990 George Miller, a medical educator, published a single diagram that has shaped assessment ever since. He was answering a question every examiner faces: *what does this test actually prove?*

His answer was a pyramid of four tiers, each demanding evidence more authentic than the one below.

- **Knows** — the student can recall it.
- **Knows how** — the student can reason with it, apply it to a case on paper.
- **Shows how** — the student can do it when someone is watching, in a set-up arranged for the purpose.
- **Does** — the student does it in real practice, unprompted, when nobody is checking.

The tiers are cumulative but not interchangeable. A student who reaches *Does* necessarily passed through the others; a student who has only been tested at *Knows* has been told nothing about the tiers above. That asymmetry is the whole point. **Evidence from a lower tier never certifies a higher one.**

WHAT THE PYRAMID IS NOT

It is not a taxonomy of thinking. It says nothing about whether a task is easy or hard, or whether it calls for recall or judgement. A *Knows* objective can be intellectually demanding and a *Does* objective can be routine. Bloom's taxonomy answers "what kind of thinking is this?"; Miller answers "how authentic must the evidence be?" The two are independent axes, and this curriculum carries both — Bloom in one column, Miller in another. Each has its own companion volume.

It is also not a ranking of worth. A curriculum that stops at *Shows how* for a dangerous procedure is not being lazy; it is making a judgement about supervision and safety. Reading the ceilings as a scoreboard is the commonest misuse of the pyramid.

> **The Cruess addition.** In 2016 Richard and Sylvia Cruess proposed a fifth tier above *Does*: **Is** — professional identity. Not what the student does but who they have become. It cannot be examined in a hall, and it is not coded anywhere in your curriculum. It is worth naming because some of what you certify at *Does* — a safety briefing recorded honestly, a result reported when it was inconvenient — is really evidence of *Is*. Where a unit in Part B says so, that is what it means.

WHY IT MATTERS HERE

These four semesters sit higher on the pyramid than anything before them: 289 of the 577 objectives — exactly half — are at *Shows how* or *Does*, and 67 reach *Does*. Half of what this course certifies cannot be certified on paper, and the summative examination at the end of Semester VI examines two years at once. Whatever was seen but not recorded is gone by then.

PART A · CHAPTER 2

The pyramid is already inside your curriculum

Miller is not something this book adds to your curriculum. It has been there since publication, in a column most teachers have never had pointed out.

COLUMN 3F

Open Table 3. The column headed **Level** carries one of four values against every objective: *Know*, *Knows how*, *Shows how* or *Does*. Those are Miller's four tiers, and they are the curriculum's statement of how authentic the evidence must be before that objective can be called met.

Across the 577 objectives of these four semesters the distribution is:

TIER	OBJECTIVES	SHARE
Knows	21	3.6%
Knows how	267	46.3%
Shows how	222	38.5%
Does	67	11.6%

Total 577 objectives.

The shape is a postgraduate one. Recall has largely been left behind — only 21 objectives sit at *Knows* — and the weight has moved to reasoning and performance. The *Does* tier is substantial rather than token, which means the record book and log book are load-bearing, not administrative.

> **A word about the other column headed "Domain".** Beside the Level column sits another carrying codes like CK, CC, CAP, PSY-MEC and AFT-VAL. Those are Bloom's taxonomy and the psychomotor and affective domains — a different question entirely, and the subject of the companion volume. In *this* book, "level" and "tier" always mean Miller.

THE CEILING

Every module in this curriculum has a **ceiling**: the highest tier any of its objectives reaches. That single fact governs assessment more than anything else on the page. It tells you what evidence you must gather, and it tells you what you may not ask for. Part B prints it for every module.

PART A · CHAPTER 3

Constructive alignment — the three rules

Constructive alignment is the principle that what you teach, what you ask students to do, and what you accept as proof must all sit at the same tier. Stated as rules, it comes to three.

RULE 1 — THE TIER DECIDES THE INSTRUMENT

Not the topic, not the hours, not the marks. Once you know an objective sits at *Shows how*, the set of instruments that can certify it is fixed, and a written paper is not among them. Chapter 4 gives the toolkit tier by tier.

RULE 2 — NEVER ASSESS ABOVE THE CEILING

If the curriculum certifies a module to *Knows how*, do not build an OSPE station for it. You would be examining something the students were never taught to that standard, and the marks would measure prior exposure rather than your teaching. This rule protects students.

RULE 3 — NEVER CERTIFY A HIGHER TIER WITH LOWER-TIER EVIDENCE

This is the rule that gets broken, and it is broken in one direction. A *Does* objective assessed by viva has not been assessed. A *Shows how* objective assessed by a long answer has not been assessed. The student may pass, the record may look complete, and nothing has been established.

> **The asymmetry is the thing to remember.** You may always gather evidence from a tier *above* the one required — a student observed doing the thing has plainly also shown how. You may never accept evidence from a tier *below*. Every alignment error worth catching is a violation of that one sentence.

WORKING THE THREE RULES BACKWARDS

In practice the fastest way to plan a session is to start at the end. Ask what evidence you will hold at the finish — a script, a checklist, a log entry. That names the tier. The tier names the instrument, and the instrument names the teaching that must precede it. A session planned forwards from the syllabus tends to arrive at the assessment it can most easily set, which is almost always a written one.

PART A · CHAPTER 4

The assessment toolkit, tier by tier

Each tier admits a particular set of instruments. Using one from the wrong tier is the single commonest fault in assessment design, and it is invisible unless you look for it, because the paperwork is complete either way.

TIER	THE QUESTION IT ANSWERS	TEACHING THAT BUILDS IT	INSTRUMENTS THAT CERTIFY IT	EVIDENCE YOU KEEP
Knows	**Can the student recall it?**	Lecture, reading, recitation, retrieval practice.	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	**Can the student reason with it?**	Worked cases, discussion, problem-based work; the student must reason aloud and be corrected.	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	**Can the student do it when watched?**	Demonstration, then supervised practice with your help withdrawn a little at a time.	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	**Does the student do it unprompted, in real practice?**	Real work with real consequences, supervised at a distance rather than at the elbow.	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

THE TRAP AT EACH TIER

- **Knows** — Spending long-answer marks here. A long answer cannot separate a student who knows the list from one who knows it better.
- **Knows how** — Setting a question answerable from memory. If the stem can be met by recall, the objective has silently dropped to Knows.
- **Shows how** — Certifying by viva. A fluent student who has never handled the material will pass a viva comfortably.
- **Does** — Certifying with a written test. Nothing written can show that a student does something unprompted when nobody is watching.

> **The instruments this curriculum names.** This curriculum sets a modular assessment in every module, worth 25 marks per credit and 1600 marks across the four semesters, and a university examination at the end of Semester VI: four papers of 100 marks and a practical examination of 400.

ON THE LOG BOOK AND RECORD BOOK

These are the only instruments in routine use that can carry *Does*-tier evidence, and they are usually treated as paperwork. A log filled in at the end of term is a piece of writing. The same log filled in as the work happens, dated and countersigned, is the only proof you will have that a student did something unprompted. Decide at the start of each module what artefact each *Does* objective will leave behind, and the record becomes evidence rather than an administrative task.

PART A · CHAPTER 5

How to use a module in Part B

Part B has thirty-two modules, each carrying its units inside it. Each begins on its own page and is laid out the same way, so once you have read one you can use any of them.

WHAT IS ON THE PAGE

1. **The heading and counters** — hours, objectives, marks, and the **ceiling**: the highest tier this module reaches. Read the ceiling first; it governs everything else.
2. **The session map** — the curriculum's own objectives with the tier each is rated to, so you can see at a glance where the weight sits.
3. **The pyramid for this module** — four rows, one per tier. Each names what a student must be able to do at that tier here, then how to build it, how to assess it, and what evidence you keep. Tiers the curriculum does not use are shown greyed, because knowing what is *not* certified matters as much as knowing what is.
4. **The ceiling statement** — a single line naming the highest tier and warning against assessing above it or certifying it with the wrong instrument.
5. **Two panels** — one for the teacher, one to give to students.

THREE WAYS TO USE IT

Planning a session. Start at the ceiling row of the pyramid and work down: that row tells you what the students must end up able to do, and the rows beneath tell you what has to be in place first.

Setting a paper. A written paper can honestly reach *Knows* and *Knows how* and no further. Use the session map to see which objectives those are; anything above must be certified elsewhere, and the module says where.

Auditing what you already do. Take your existing assessment for a module and ask which tier it can certify. If it is a written test and the ceiling is *Shows how* or *Does*, the gap is your finding.

> **What this book does not do.** It does not rewrite the curriculum and does not tell you what to teach — every objective in Part B appears in the curriculum's own words. What it adds is the Miller layer: what standard of evidence each objective demands, and which instruments can supply it.

PART B

The modules

Thirty-two modules, ordered by semester and then by paper so the modules a department teaches together sit together. Each carries its units inside it, with every objective gathered under the tier its Level column gives it and the kind of hour beside it. The pyramid block names what every tier asks for in that module, how to build it, how to assess it and what evidence to keep. Chapter 5 explains the layout.

1. **Semester III** — Modules 1, 2, 9, 10, 17, 18, 25, 26
2. **Semester IV** — Modules 3, 4, 11, 12, 19, 20, 27, 28
3. **Semester V** — Modules 5, 6, 13, 14, 21, 22, 29, 30
4. **Semester VI** — Modules 7, 8, 15, 16, 23, 24, 31, 32

↑ Contents

MODULE 1 · SEMESTER 3 · PAPER 1

Chronological Development and Taxonomy of Rasa Dravyas

2 credits | 60 notional hours | 4 units | **19 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	10	7	2
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Recognise Rasadravya based on their Graahyatva-Agraahyatva; Identify Rasadravya using geological/mineralogical characters.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Construct a chronological tree of Rasashastra and its integration with other medical traditions; Demonstrate the application of the Mendeleev Periodic table in mapping Rasadravyas; Identify and analyse selected metals and minerals with the periodic table elements; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
10 objectives

Discuss the philosophical relevance of Rasashastra and historical evidence till the 6th century AD; Describe the chronological development of Rasashastra from the 6th Century AD to the medieval period; Discuss the chronological development of Rasashastra after the medieval period; and 7 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Chronological Development and Overview of Rasashastra

5 obj · 9 h · ceiling Shows how

1.1 Philosophical Relevance of Rasashastra
1.2 Chronological Development of Rasashastra
1.3 Integration of the development of Iatrochemistry with modern chemistry

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Discuss the philosophical relevance of Rasashastra and historical evidence till the 6th century AD.	Lecture	1	Knows how
Describe the chronological development of Rasashastra from the 6th Century AD to the medieval period.	Lecture	1	Knows how
Discuss the chronological development of Rasashastra after the medieval period.	Lecture	1	Knows how
Integrate the development of Iatrochemistry with modern chemistry.	Lecture	1	Knows how
SHOWS HOW — 1			
Construct a chronological tree of Rasashastra and its integration with other medical traditions.	Experiential Learning	5	Shows how

Unit 2 · Taxonomy of Rasadravyas

5 obj · 13 h · ceiling Shows how

2.1 Taxonomy of Rasadravyas
2.2 Concepts and justification about division of Rasadravyas
2.3 Mendeleeve periodic table

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Elaborate classification of Rasadravyas as per different ancient classics.	Lecture	1	Knows how
Justify the division of Rasadravyas.	Lecture	1	Knows how
Discuss the critics of the classification of Rasadravya.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the application of the Mendeleev Periodic table in mapping Rasadravyas.	Practical Training	5	Shows how
Identify and analyse selected metals and minerals with the periodic table elements.	Experiential Learning	5	Shows how

Unit 3 · Geological/mineralogical identification of Rasadravyas

6 obj · 22 h · ceiling Does

3.1 Grahya-Agrahyata of Rasadravyas
3.2 Geological/Mineralogical identification of Rasadravyas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss classical parameters for differentiating acceptable and non-acceptable Rasadravya.	Lecture	1	Knows how
Apply geological/mineralogical characters for the identification of Rasadravya.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the comparison and analyze the classification of Rasa Dravyas.	Practical Training	5	Shows how
Demonstrate Geological/Mineralogical identification of Rasadravyas.	Practical Training	5	Shows how
DOES — 2			
Recognise Rasadravya based on their Graahyatva-Agraahyatva.	Experiential Learning	5	Does
Identify Rasadravya using geological/mineralogical characters.	Experiential Learning	5	Does

Unit 4 · Overview of Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya

3 obj · 12 h · ceiling Shows how

4.1 Sandigdha (Controversial)
4.2 Anupalabdha (Not Available) and Krutrima (Artificial) Rasadravya
4.3 Pratinidhi (Substitute) Rasadravya

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstratie the Krutrima and Pratinidhi Rasadravyas.	Practical Training	5	Shows how
Identify and analyse Sandigdha and Anuplabhdha Rasadravyas and propose suitable alternatives.	Experiential Learning	6	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **9 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 2 · SEMESTER 3 · PAPER 1

Puta and Samskara (Transformative processes of drug manufacturing)

2 credits | 60 notional hours | 6 units | **21 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	7	11	1
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	9	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	8	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Explore Thermal Dynamics in Traditional Puta Systems: An of Fuel Variations, Temperature Patterns, and Thermodynamic Principles.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
11 objectives

Demonstrate and compare the temperature pattern and total duration of different traditional Puta with EMF; Perform comparison of Traditional and Modern Puta Methods: Observing and Analyzing Temperature Patterns and Duration; Demonstrate the art of Gandhaka Jarana: A Process, Understanding its Role and Significance in Rasashastra; and 8 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
7 objectives

Explore the traditional concept of Puta, discussing its definition, significance, and classification in the context of Ayurvedic pharmaceutical practices; Explain the laws of thermodynamics (Zeroth, First, Second, Third), the law of thermochemistry, Griffith's theory, and Collision theory; Discuss the fundamental principles and significance of Jarana, Amritikarana, and Lohitikarana procedures in the preparation of bhasmas; and 4 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Examine the modern applications and relevance of Puta in contemporary pharmaceutical processing, integrating traditional knowledge with modern technology; Discuss the concepts of Kalaprakarsha (processing with specific time exposure) and Bhajana (processing vessels) in Ayurvedic pharmaceutical processing, including the significance of processing time, vessel material, and other temporal factors on drug quality and efficacy.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Puta and Law of thermodynamics

6 obj · 18 h · ceiling Does

1.1 Definition, Significance and Classification of Puta

1.2 Laws of thermodynamics, Thermochemistry, Griffith theory, Collision theory, etc

1.3 Heating devices and fuels, with detailed knowledge about temperature, duration, designing and construction of different Puta

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Examine the modern applications and relevance of Puta in contemporary pharmaceutical processing, integrating traditional knowledge with modern technology.	Lecture	1	Knows
KNOWS HOW — 2			
Explore the traditional concept of Puta, discussing its definition, significance, and classification in the context of Ayurvedic pharmaceutical practices.	Lecture	1	Knows how
Explain the laws of thermodynamics (Zeroth, First, Second, Third), the law of thermochemistry, Griffith's theory, and Collision theory.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate and compare the temperature pattern and total duration of different traditional Puta with EMF.	Practical Training	5	Shows how
Perform comparison of Traditional and Modern Puta Methods: Observing and Analyzing Temperature Patterns and Duration.	Experiential Learning	5	Shows how
DOES — 1			
Explore Thermal Dynamics in Traditional Puta Systems: An of Fuel Variations, Temperature Patterns, and Thermodynamic Principles.	Experiential Learning	5	Does

Unit 2 · Concept, understanding, application, SOPs and chemical changes of Jarana, Amritikarana and Lohitikarana

4 obj · 16 h · ceiling Shows how

2.1 Principle and Significance of

2.2 Application of Amritikarana and Lohitikarana procedures in bhasma preparation

2.3 Standard Operating Procedures (SOP) and Chemical Changes of Jarana, Amritikarana, and Lohitikarana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss the fundamental principles and significance of Jarana, Amritikarana, and Lohitikarana procedures in the preparation of bhasmas.	Lecture	2	Knows how
SHOWS HOW — 3			
Demonstrate the art of Gandhaka Jarana: A Process, Understanding its Role and Significance in Rasashastra.	Practical Training	5	Shows how
Demonstrate Amritikarana and Lohitikarana: Traditional Methods for Mica Processing in Rasashastra.	Practical Training	5	Shows how
Perform an experiential Investigation of Amritikarana and Lohitikarana Procedures Exploring Safety, Efficacy, and Principles of Transformation.	Experiential Learning	4	Shows how

Unit 3 · Concept, understanding, application, SOPs (Standard Operating Procedures) and chemical changes of Satva and Druti procedures

4 obj · 9 h · ceiling Shows how

3.2 Application and analysis of Satva and Druti procedures

3.3 SOP and chemical changes of Satva and Druti

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the principles, significance, and applications of Satva procedures within the context of Ayurvedic pharmacy and modern pharmaceutical practices.	Lecture	1	Knows how
Describe the principles, significance, and apply Druti procedures within the context of Ayurvedic pharmacy.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the procedure of Satvapatana.	Practical Training	5	Shows how
Perform and analyse the Satva Patana procedure.	Experiential Learning	2	Shows how

Unit 4 · Samskara (Principles of Transformative processes of drug manufacturing)

2 obj · 3 h · ceiling Shows how

4.1 Application of Toya Sannikarsha (Processing with water) and Agni

4.2 Shaucha and Manthana (churning)

4.3 Vasana (process of infusing aroma), and Bhavana (Effect of Levigation / Trituration)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Integrate and apply classical Ayurvedic manufacturing principles, including water, heat and combined processing, cleaning, churning, proximity, and trituration effects.	Lecture	1	Knows how
SHOWS HOW — 1			
Apply Time-Tested Principles of Classical Drug Manufacturing: by integrating Toya Sannikarsha, Agni Sannikarsha, Toyagni Sannikarsha, Shaucha, and Manthana to Optimize Pharmaceutical Quality, Safety, and Efficacy.	Experiential Learning	2	Shows how

Unit 5 · Influence of Desha (place) and Kala(time and seasonal factors) of drug qualities

2 obj · 3 h · ceiling Shows how

5.1 Influence of Desha (geographical area and place of

5.2 Influence of Kala (processing time, season, and maturity of herbs) on drug quality

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Description of the influence of Desha (geographical area) and Kala (time and seasonal factors) on the quality of Ayurvedic drugs, including the impact of geographical location, climate, and seasonal variations on drug efficacy and safety.	Lecture	1	Knows how
SHOWS HOW — 1			
Perform A Hands-on Investigation of Geographical and Temporal Influences on drug 7, CO 8 quality: Desha–Kala in Pharmaceutical Manufacturing.	Experiential Learning	2	Shows how

Unit 6 · Kalaprakarsha- Processing with/Effect of specific time exposure and Bhajana (processing vessel)

3 obj · 7 h · ceiling Shows how

5.1 Knowledge of Kalaprakarsha (Time taken

5.2 Knowledge of Bhajana (vessels used for manufacturing)

5.3 Environmental and Temporal Influences on Drug Quality and Processing

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss the concepts of Kalaprakarsha (processing with specific time exposure) and Bhajana (processing vessels) in Ayurvedic pharmaceutical processing, including the significance of processing time, vessel material, and other temporal factors on drug quality and efficacy.	Lecture	1	Knows

SHOWS HOW — 2			
Demonstrate and analyze the role of Vasana and Bhavana to Optimize Pharmaceutical Quality, Safety, and Efficacy.	Experiential Learning	4	Shows how
Demonstrate the effect of Kalaprakarsha and Bhajana in Pharmaceutical Processing Analyzing Processes and Findings to Optimize Quality, and stability.	Experiential Learning	2	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **12 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 9 · SEMESTER 3 · PAPER 2

Scientific fundamentals of Ayurvedic Pharmaceutical preparations

2 credits | 60 notional hours | 4 units | **15 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	5	6	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Demonstrate processing of unique Panchavidha Kashaya Kalpana with their ingredients proportion of water and procedural details; Review Research Updates in the field of Ayurvedic pharmaceutics along with modern scientific validation methods.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Prepare unique formulation from the category of Panchavidha Kashaya Kalpana; Identify and apply various tools and techniques used for quality control of raw materials and appraise the classical rules of the collection of raw material; Demonstrate various formulations by describing their ingredients and understanding the uniqueness of their proportion; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Discuss the rationality behind the selection of Panchavidha Kashaya Kalpana and compare classical and modern drug extraction techniques; Describe classical and contemporary parameters for quality assurance of raw drugs, as well as adulteration and contamination of the raw material; Demonstrate application of QAQC parameters used for quality control of herbal drugs; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Describe the classical and contemporary methods used for the processing of drugs including the assessment of the End-Point(Siddhi Lakshanani) of the dosage form, C_L; Recognize the factors responsible for a change in the concentration of the prepared drug and assess the extracted phytochemicals with the help of TLC and other methods of analysis.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Classical Principles of drug processing and scientific basis of Panchavidha Kashaya Kalpana

3 obj · 6 h · ceiling Does

1.1 Bhaishajya Synonyms and significance of its
1.2 Classical principles of processing
1.3 Scientific review of Panchavidha Kashaya Kalpana and Upakalpana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss the rationality behind the selection of Panchavidha Kashaya Kalpana and compare classical and modern drug extraction techniques.	Lecture	1	Knows how
SHOWS HOW — 1			
Prepare unique formulation from the category of Panchavidha Kashaya Kalpana.	Experiential Learning	3	Shows how
DOES — 1			
Demonstrate processing of unique Panchavidha Kashaya Kalpana with their ingredients proportion of water and procedural details.	Practical Training	2	Does

Unit 2 · Identity, Purity and Quality of Raw materials used in Ayurveda Pharmaceutics

3 obj · 12 h · ceiling Shows how

2.1 Classical knowledge about Identity, Purity, Quality of Raw
2.2 Collection of Raw material- GACP, QAQC parameters
2.3 Adulteration and Contamination
2.4 Substitute drugs (Pratinidhi Dravya)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe classical and contemporary parameters for quality assurance of raw drugs, as well as adulteration and contamination of the raw material.	Lecture	2	Knows how
Demonstrate application of QAQC parameters used for quality control of herbal drugs.	Practical Training	4	Knows how
SHOWS HOW — 1			
Identify and apply various tools and techniques used for quality control of raw materials and appraise the classical rules of the collection of raw material.	Experiential Learning	6	Shows how

Unit 3 · Pharmaceutical processing of drugs with its rationality

3 obj · 13 h · ceiling Shows how

3.1 Processing of Drugs- Classical methodology
3.2 Simple and unique remedies from classical texts prepared with various processing methods
3.3 Rationality behind the Processing Method of Classical Dosage Forms
3.4 Siddhilakshanani (End-point assessment of pharmaceutical process) and its significance

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the classical and contemporary methods used for the processing of drugs including the assessment of the End-Point(Siddhi Lakshanani) of the dosage form, C_L.	Lecture	2	Knows
KNOWS HOW — 1			
Interpret significance of understanding the proportion of ingredients of formulations quoted in the classics.	Experiential Learning	6	Knows how
SHOWS HOW — 1			
Demonstrate various formulations by describing their ingredients and understanding the uniqueness of their proportion.	Practical Training	5	Shows how

Unit 4 · Ayurvedic Classical Dosage Forms and their integration with modern scientific validation

6 obj · 25 h · ceiling Does

4.1 Classification of the source of raw material and the
4.2 Multiple factors responsible for Process optimization
4.3 Key qualities of good pharmaceutical products to ensure safety, efficacy, and authenticity

4.4 Research updates about the Integration of traditional wisdom with modern scientific validation

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Recognize the factors responsible for a change in the concentration of the prepared drug and assess the extracted phytochemicals with the help of TLC and other methods of analysis.	Lecture	3	Knows
KNOWS HOW — 1			
Describe multiple factors responsible for the process optimization of Ayurvedic pharmaceutical preparations and illustrate key qualities of good pharmaceutical products to ensure safety, efficacy, and authenticity.	Lecture	2	Knows how
SHOWS HOW — 3			
Demonstrate the extraction of phytochemicals in the prepared dosage form.	Practical Training	4	Shows how
Perform the process of Extraction of Phytochemicals in Ayurvedic formulations.	Experiential Learning	5	Shows how
Demonstrate the kwatha preparation considering its process optimization.	Practical Training	5	Shows how
DOES — 1			
Review Research Updates in the field of Ayurvedic pharmaceutics along with modern scientific validation methods.	Experiential Learning	6	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 10 · SEMESTER 3 · PAPER 2

Principles of pharmaceutical formulation

2 credits | 60 notional hours | 5 units | **12 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	6	5	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Justify the need for Shaktyutkarsha (Potency enhancement)in drug preparation and Advocate techniques of potency enhancement used in Ayurvedic pharmaceutics.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
5 objectives

Illustrate the role of samskaras in the changes observed during the preparation of various dosage forms according to classical texts; Demonstrate combinations of various classical formulations and justify the rationale behind the ingredients; Demonstrate various designs of dosage forms, considering all fundamentals of pharmacology; and 2 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
6 objectives

Describe the applied principles of samskara involved in the preparation of various kalpana; Analyze the purpose of composition and combinations of ingredients in classical formulations; Justify the Principles of dosage form design as per modern pharmaceutical science; and 3 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Principles of Samskara (transformative processes of drug manufacturing) in relation to drug development

2 obj · 6 h · ceiling Shows how

1.1 Qualities of drugs/Procedures
1.2 Classical concept of Samskara and expected alterations in qualities
1.3 Key role of Samskara in pharmacological and therapeutic modification of the drug
1.4 Modifications in samskaras in current practices and their consequences

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe the applied principles of samskara involved in the preparation of various kalpana.	Lecture	2	Knows how
SHOWS HOW — 1			
Illustrate the role of samskaras in the changes observed during the preparation of various dosage forms according to classical texts.	Practical Training	4	Shows how

Unit 2 · Rationality behind Composition of Formulations

2 obj · 6 h · ceiling Shows how

2.1 Combinations of ingredients (desirable/ inadmissible)
2.2 Rational integrity behind the combination of ingredients
2.3 Aushadhi Gana (therapeutic groups) or Varga (categories) of medicinal herbs and their role in formulations
2.4 Combinations based on predominant pharmacological actions, qualities, or therapeutic usage of the ingredients

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Analyze the purpose of composition and combinations of ingredients in classical formulations.	Lecture	2	Knows how
SHOWS HOW — 1			
Demonstrate combinations of various classical formulations and justify the rationale behind the ingredients.	Practical Training	4	Shows how

Unit 3 · Principles and designing of various dosage forms

2 obj · 8 h · ceiling Shows how

3.1 Principles of dosage form design
3.2 Pharmaceutical Considerations
3.3 Formulation Considerations
3.4 Biopharmaceutical Considerations
3.5 Pharmacokinetic Considerations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Justify the Principles of dosage form design as per modern pharmaceutical science.	Lecture	2	Knows how
SHOWS HOW — 1			
Demonstrate various designs of dosage forms, considering all fundamentals of pharmacology.	Practical Training	6	Shows how

Unit 4 · Principles of Prakshep dravyas (Pharmaceutical Additives)

3 obj · 14 h · ceiling Shows how

4.1 Prakshepa Dravya- Need, Importance, Proportion (matra)
4.2 Pharmaceutical Excipients

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Identify the selection and relevance of Prakshepa dravya added to various dosage forms and describe Pharmaceutical excipients.	Lecture	2	Knows how

SHOWS HOW — 2			
Demonstrate and analyze the importance of pharmaceutical excipients.	Practical Training	6	Shows how
Illustrate various Prakshepa Dravyas and their proportions stated in the classics.	Experiential Learning	6	Shows how

Unit 5 · Principles of Potency and Efficacy enhancement

3 obj · 22 h · ceiling Does

5.1 Classical principles of Potency and efficacy enhancement (Shaktyutkarsha)
5.2 Methods of Potency and Efficacy Enhancement in Ayurvedic Classics
5.3 Methods of Potency and Efficacy Enhancement in Modern Pharmaceutics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe and compare various techniques used in the classics for Shaktyutkarsha (Potency enhancement).	Lecture	2	Knows how
Investigate the principles for improving potency and efficacy in modern pharmaceutics.	Experiential Learning	10	Knows how
DOES — 1			
Justify the need for Shaktyutkarsha (Potency enhancement)in drug preparation and Advocate techniques of potency enhancement used in Ayurvedic pharmaceutics.	Experiential Learning	10	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **6 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 17 · SEMESTER 3 · PAPER 3

Guidelines for administration of Rasoushadhi in clinical Practice

2 credits | 60 notional hours | 5 units | **22 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	10	9	3
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	7	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Generate document depicting the outcome of compliance and non-compliance with the pathya apathya guidelines; Evaluate the prescription errors containing Rasoushadhis; Evalute the effect of your chelation therapy in the management of our heavy metal toxicity.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
9 objectives

Demonstrate the rational prescription for a selected disease; Apply the principles of Rasa Chikitsa in the clinical practice/ alloted case; Demonstrate the need and role of Kshetrikarana; and 6 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
10 objectives

Analyse the term Rasa from the Sanskrit and Ayurvedic literatures and discuss the significance of Rasoushadhi; Discuss various treatment modalities used in the Ayurvedic system of medicine; Define kshetrikarana and discuss the concept of kshetrikarana; and 7 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Concept of Rasa chikitsa and significance of Rasoushadhi

4 obj · 11 h · ceiling Shows how

1.1 Perspectives on the Notion of Rasa
1.2 Significance of Rasa Chikitsa
1.3 Various treatment modalities of Ayurveda

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Analyse the term Rasa from the Sanskrit and Ayurvedic literatures and discuss the significance of Rasoushadhi.	Lecture	1	Knows how
Discuss various treatment modalities used in the Ayurvedic system of medicine.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the rational prescription for a selected disease.	Practical Training	3	Shows how
Apply the principles of Rasa Chikitsa in the clinical practice/ alloted case.	Experiential Learning	6	Shows how

Unit 2 · Concept of Kshetrikarana and its significance prior to administration of Rasoushadhi

4 obj · 11 h · ceiling Shows how

2.1 Etymology and Concept of Kshetrikarana
2.2 Kshetrikarana for Optimising Medicine Efficacy
2.3 Procedure of Kshetrikarana (Panchakarma & Pancha Shodhana)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Define kshetrikarana and discuss the concept of kshetrikarana.	Lecture	1	Knows how
Discuss the need and benefits of Kshetrikarana.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the need and role of Kshetrikarana.	Practical Training	3	Shows how
Demonstrate the procedure and benefits of kshetrikarana.	Experiential Learning	6	Shows how

Unit 3 · Pathya -Apathya during Rasayoga Sevana

5 obj · 12 h · ceiling Does

3.1 Pathya and Apathya Ahara, Vihara, and Manasika Vikara during Rasayoga Sevana
3.2 Importance of Pathya-Apathya during Rasayoga Sevana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Define Pathya and Apathya ahara and describe its importance.	Lecture	1	Knows how
Describe the application of the concept of what pathya and apathya vihara with respect to the Manasika vikara.	Lecture	1	Knows how
SHOWS HOW — 2			
Enlist and demonstrate common pathya and apathya Ahara during rasa yoga sevana.	Practical Training	2	Shows how
Enlist and demonstrate the common pathya-apathya Vihara during rasa yoga sevana.	Practical Training	2	Shows how
DOES — 1			
Generate document depicting the outcome of compliance and non-compliance with the pathya apathya guidelines.	Experiential Learning	6	Does

Unit 4 · Safety and toxicity of Rasoushadhi

5 obj · 14 h · ceiling Does

4.1 Safety considerations of Rasoushadhi
4.2 Ashuddha and Apakwa Bhasma Sevana Janya Vikara and their management

4.3 Safety and prescription trends of Rasoushadhi

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the safety considerations of Rasa oushadhies in detail.	Lecture	1	Knows how
Analyse the consequences of Ashuddha or Apakwa Bhasma sevana Dosha, along with the treatment.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate a case study of signs and symptoms of rasa vikaras and their management.	Practical Training	6	Shows how
Assess and comapre Ashuddha and Shuddha Bhasma effects in the clinical practice.	Practical Training	2	Shows how
DOES — 1			
Evaluate the prescription errors containing Rasoushadhis.	Experiential Learning	4	Does

Unit 5 · Heavy metal toxicity and chelation therapy

4 obj · 8 h · ceiling Does

5.1 The mechanism of heavy metal toxicity
5.2 Symptoms of heavy metal toxicity
5.3 Factors influencing heavy metal toxicity
5.4 Concept of chelation therapy and risks, and benefits

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss in detail the sources, pathways, and factors influencing heavy metal toxicity.	Lecture	1	Knows how
Discuss chelating agents and their role in managing heavy metal toxicity.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate identification of heavy metal toxicity.	Practical Training	2	Shows how
DOES — 1			
Evalute the effect of your chelation therapy in the management of our heavy metal toxicity.	Experiential Learning	4	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **12 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 18 · SEMESTER 3 · PAPER 3

Prayogika Bhaishajya Kalpana(Clinical application of Pharmaceutical products)

2 credits | 60 notional hours | 4 units | **24 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	11	11	2
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	8	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Demonstrate Modulation of dosage form action by chronotherapeutic administration in Ayurveda; Describe the role of Anupana in the modulation of drug tolerance and safety.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
11 objectives

Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics; Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics; Illustrate the modulation of the action of the dosage form by chronotherapeutic administration in Ayurveda; and 8 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
11 objectives

Discuss in detail the concept of Oushadha Sevana Kala in Ayurveda; Discuss biological rhythms and chronobiology and their significance in drug delivery; Elaborate the principles of chrono pharmaceutical drug delivery systems and their benefits; and 8 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Aushadha Sevana Kala (Time of administration of drug)

7 obj · 15 h · ceiling Does

1.1 Classical concept of time of administration
1.2 Biological rhythms and chronobiology
1.3 Chrono-pharmaceutics and chronopharmaceutical drug delivery systems
1.4 Principles of chronopharmacology

1.5 Benefits of chronotherapeutics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss in detail the concept of Oushadha Sevana Kala in Ayurveda.	Lecture	1	Knows how
Discuss biological rhythms and chronobiology and their significance in drug delivery.	Lecture	1	Knows how
Elaborate the principles of chrono pharmaceutical drug delivery systems and their benefits.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics.	Practical Training	3	Shows how
Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics.	Experiential Learning	3	Shows how
Illustrate the modulation of the action of the dosage form by chronotherapeutic administration in Ayurveda.	Experiential Learning	3	Shows how
DOES — 1			
Demonstrate Modulation of dosage form action by chronotherapeutic administration in Ayurveda.	Practical Training	3	Does

Unit 2 · Concept and scientific basis of Anupana

6 obj · 16 h · ceiling Does

2.1 Classical concepts of Anupana in Ayurveda
2.2 Role of Anupana in bioavailability
2.3 Anupana for targeted drug delivery and distribution
2.4 Role of Anupana in the mitigation of adverse effects and toxicity
2.5 Anupana for synergistic effect

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss in detail the concept of anupana in Ayurveda.	Lecture	1	Knows how
Explain the role of Anupana in bioavailability and targeted drug delivery.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate appropriate Anupana for the selected dosage form in different clinical conditions.	Practical Training	3	Shows how
Enlist appropriate Anupanas for various formulations in different clinical conditions.	Experiential Learning	3	Shows how
Demonstrate drug response through synergistic Anupana integration.	Experiential Learning	4	Shows how
DOES — 1			
Describe the role of Anupana in the modulation of drug tolerance and safety.	Experiential Learning	4	Does

Unit 3 · Matra (Posology)

5 obj · 12 h · ceiling Shows how

3.1 Definition, importance, and clinical significance
3.2 Factors affecting drug dose
3.3 Posology according to modern pharmacology
3.4 Calculation of paediatric dose

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Elaborate on the concept, importance, and clinical significance of Matra.	Lecture	1	Knows how
Discuss the factors affecting drug dose in detail.	Lecture	1	Knows how
Demonstrate the calculation of paediatric dose using different formulae.	Practical Training	3	Knows how

SHOWS HOW — 2			
Demonstrate the Patient-centric evaluation of Matra.	Practical Training	3	Shows how
Demonstrate the utility of posology principles from an ayurvedic and modern pharmacology perspective.	Experiential Learning	4	Shows how

Unit 4 · Concept of drug repurposing

6 obj · 13 h · ceiling Shows how

4.1 Concept of drug repurposing
4.2 Systematic approaches to drug repurposing
4.3 Strategies of drug repurposing
4.4 Methodologies of drug repurposing

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe the concept, need, and key aspects of drug repurposing.	Lecture	1	Knows how
Describe systematic approaches to drug repurposing.	Lecture	1	Knows how
Discuss strategies and methodologies of drug repurposing.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the analysis of classical ayurvedic drugs for novel indications.	Practical Training	5	Shows how
Evaluate novel indications for classical Ayurvedic formulations.	Experiential Learning	3	Shows how
Demonstrate the Reverse pharmacological approach for drug repurposing in Ayurveda.	Experiential Learning	2	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **13 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 25 · SEMESTER 3 · PAPER 4

Artificial intelligence in drug development and Pharmacogenetics; Pharmacogenomics

2 credits | 60 notional hours | 5 units | **20 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	11	7	2
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Perform De Novo Drug. Design Using Generative Models; Demonstrate prediction of drug likeliness using machine learning.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate the basics of machine learning and its application; Demonstrate different 3D printers used in the drug delivery system; Analyse ADMET properties using AI tools; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
11 objectives

Define AI and discuss its importance in drug development; Discuss the Role of AI in Drug Discovery and Drug development with a suitable example; Describe different types of software used in AI; and 8 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Role of Artificial intelligence in drug development

4 obj · 11 h · ceiling Does

1.1 Introduction to AI in drug development
1.2 AI, machine-based intelligence learning
1.3 Role of AI in Drug discovery and Drug development

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Define AI and discuss its importance in drug development.	Lecture	1	Knows how
Discuss the Role of AI in Drug Discovery and Drug development with a suitable example.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the basics of machine learning and its application.	Practical Training	4	Shows how
DOES — 1			
Perform De Novo Drug. Design Using Generative Models.	Experiential Learning	5	Does

Unit 2 · Types of software used in Artificial intelligence

3 obj · 6 h · ceiling Does

2.1 Different types of software used in AI
2.2 Data mining and analysing medical records, medication management
2.3 Recent updates

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe different types of software used in AI.	Lecture	1	Knows how
Discuss Data mining and analysing medical records, medication management, with recent updates.	Lecture	1	Knows how
DOES — 1			
Demonstrate prediction of drug likeliness using machine learning.	Practical Training	4	Does

Unit 3 · 3D printed drug delivery system

5 obj · 16 h · ceiling Shows how

3.1 Approaches of 3D printing in current drug delivery
3D printing in personalized drug delivery
3D-Printed Drug Delivery Systems Based on Natural Products

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe the fundamentals and explore the materials and technology used in 3D printing.	Lecture	1	Knows how
Discuss the concept of personalised drug delivery, formulations, and natural products.	Lecture	1	Knows how
Demonstrate the comparison of traditional dosage forms versus 3D printed ones regarding the structure and dissolution.	Practical Training	4	Knows how
SHOWS HOW — 2			
Demonstrate different 3D printers used in the drug delivery system.	Practical Training	4	Shows how
Analyse ADMET properties using AI tools.	Experiential Learning	6	Shows how

Unit 4 · Pharmacogenetics and Pharmacogenomics

4 obj · 11 h · ceiling Shows how

4.1 Introduction to Pharmacogenetics and Pharmacogenomics along with its importance

4.2 Factors influencing drug response and ADE (Adverse drug events)

4.3 Methods of pharmacogenetics testing

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Appraise key concepts and terminologies and explore the role of Pharmacogenetics and Pharmacogenomics.	Lecture	1	Knows how
Justify clinical application of pharmacogenetics, benefits, and challenges.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate to predict Drug-Target Interaction.	Practical Training	4	Shows how
Identify potential adverse effects of drug candidates using dataset and tools.	Experiential Learning	5	Shows how

Unit 5 · Pharmacogenomics in drug discovery

4 obj · 12 h · ceiling Shows how

5.1 Benefits of Pharmacogenomics

5.2 Drug Repurposing

5.3 Basis of gene-drug association

5.4 Current status and future perspective of Pharmacogenomics

5.5 Ethical concern

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss pharmacogenomics data contributes to the identification and validation of drug targets.	Lecture	1	Knows how
Discuss the current status and future perspectives of pharmacogenomics, including ethical concerns.	Lecture	1	Knows how
SHOWS HOW — 2			
Extract potential drug-disease relationship by using through Literature /data Mining.	Experiential Learning	5	Shows how
Interpret Pharmacokinetics and Toxicology Modelling using appropriate AI tools and methods.	Experiential Learning	5	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **9 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 26 · SEMESTER 3 · PAPER 4

Pharmacology

2 credits | 60 notional hours | 5 units | **16 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	11	1	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	2	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Analyse Adverse Drug Reaction in a polypharmacy patient through a case scenario; Discuss Emergency Pharmacology through a case scenario of anaphylaxis management.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Analyse the mechanism of action of drugs acting on the respiratory system through a case study scenario.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
11 objectives

Discuss dose response relationship, and factors influencing drug action; Demonstrate pharmacological experiments using videos or simulation software; Discuss therapeutic index, drug interactions, and adverse drug reactions; and 8 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Define and classify pharmacology and describe the principles and mechanisms of drug action, receptor theories, and classification of rece; Describe the basic principles of pharmacokinetics (absorption, distribution, metabolism 7, CO 8 excretion).

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to Pharmacology

3 obj · 12 h · ceiling Knows how

1.1 Introduction to Pharmacology
1.2 Pharmacodynamics
1.3 Dose-response relationship and factors influencing drug action

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Define and classify pharmacology and describe the principles and mechanisms of drug action, receptor theories, and classification of rece.	Lecture	1	Knows
KNOWS HOW — 2			
Discuss dose response relationship, and factors influencing drug action.	Lecture	1	Knows how
Demonstrate pharmacological experiments using videos or simulation software.	Practical Training	10	Knows how

Unit 2 · Pharmacokinetics, therapeutic index, and drug interactions

3 obj · 12 h · ceiling Knows how

2.1 Pharmacokinetics (absorption, distribution, metabolism, excretion)
2.2 Therapeutic index and combined effects of drugs
2.3 Adverse drug reaction
2.4 Drug interactions (pharmacokinetic and pharmacodynamic)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the basic principles of pharmacokinetics (absorption, distribution, metabolism 7, CO 8 excretion).	Lecture	1	Knows
KNOWS HOW — 2			
Discuss therapeutic index, drug interactions, and adverse drug reactions.	Lecture	1	Knows how
Demonstrate the effect of drugs on animals by simulated experiments.	Practical Training	10	Knows how

Unit 3 · Pharmacology of drugs that act on the peripheral nervous system

3 obj · 9 h · ceiling Knows how

3.1 Organization and function of ANS (Autonomic Nervous System)
3.2 Neurohumoral transmission, co-transmission, and classification of neurotransmitters
3.3 Parasympathomimetics, Parasympatholytics, Sympathomimetics, Sympatholytics
3.4 Neuromuscular blocking agents and skeletal muscle relaxants (peripheral)
3.5 Local anesthetic agents

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the organisation and function of ANS (Autonomic Nervous System) and describe Neurohumoral transmission, co-transmission, and classification of neurotransmitters.	Lecture	1	Knows how
Enlist Parasympathomimetics, Parasympatholytics, Sympathomimetics, sympatholytics Neuromuscular blocking agents, and skeletal muscle relaxants (peripheral) and Local anaesthetic agents.	Lecture	1	Knows how
Analyse the mechanism of action of drugs acting on the Cardiovascular system through a case study scenario.	Experiential Learning	7	Knows how

Unit 4 · Pharmacology of drugs that act on the central nervous system I

3 obj · 9 h · ceiling Shows how

4.1 Neurohumoral transmission in the C.N.S. special emphasis on the importance of
4.2 General anaesthetics and pre-anaesthetics
4.3 Sedatives, hypnotics, and centrally acting muscle relaxants
4.4 Anti-epileptics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe Neurohumoral transmission in the C.N.S and illustrate General anaesthetics and pre-anaesthetics.	Lecture	1	Knows how
Categorize drugs into classes such as Sedatives, hypnotics, and centrally acting muscle relaxants, and discuss anti-epileptics.	Lecture	1	Knows how
SHOWS HOW — 1			
Analyse the mechanism of action of drugs acting on the respiratory system through a case study scenario.	Experiential Learning	7	Shows how

Unit 5 · Pharmacology of drugs that act on the central nervous system II

4 obj · 14 h · ceiling Does

5.1 Psychopharmacological agents: Antipsychotics, antidepressants, anti-anxiety
5.2 Drugs used in Parkinson's disease and Alzheimer's disease
5.3 CNS stimulants and nootropics
5.4 Drug addiction, drug abuse, tolerance, and dependence

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss Psychopharmacological agents and enlist drugs used in Parkinson's disease and Alzheimer's disease.	Lecture	1	Knows how
Differentiate CNS stimulants and nootropics, and discuss drug addiction, drug abuse tolerance, and dependence.	Lecture	1	Knows how
DOES — 2			
Analyse Adverse Drug Reaction in a polypharmacy patient through a case scenario.	Experiential Learning	6	Does
Discuss Emergency Pharmacology through a case scenario of anaphylaxis management.	Experiential Learning	6	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **3 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 3 · SEMESTER 4 · PAPER 1

Parada (Mercury) and Its Processing: Ancient wisdom and Scientific advances

2 credits | 60 notional hours | 4 units | **17 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	7	7	1
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	9	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	3	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Perform the Vishesha Shodhana procedures of Parada.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate Samanya and Vishesha Shodhana procedures of Parada; Perform the Parada Jarana, Parada Murchana and Bandhana processes; Perform the Samanya Shodhana of Parada; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
7 objectives

Describe the ancient and contemporary perspectives of Parada (Mercury); Discuss the rationale behind Samanya and Vishesha Shodhana; Describe the scientific basis of Parada Jarana, Murcchana, and Bandha; and 4 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Discuss the processes related to Parada and their significance; Describe and elaborate on the metallurgy and stoichiometry of mercury.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Parada: Ancient and Contemporary Perspectives on Its Shodhana and Underlying Principles

8 obj · 22 h · ceiling Does

1.1 Ancient & Contemporary knowledge of Parada/
1.2 Parada Dosha, Shodhana(Purification) of Parada and its significance
1.3 Principles of Parada Jarana and Parada Murcchana
1.4 Utility of Parada Bandha

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss the processes related to Parada and their significance.	Lecture	1	Knows
KNOWS HOW — 3			
Describe the ancient and contemporary perspectives of Parada (Mercury).	Lecture	1	Knows how
Discuss the rationale behind Samanya and Vishesha Shodhana.	Lecture	1	Knows how
Describe the scientific basis of Parada Jarana, Murcchana, and Bandha.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate Samanya and Vishesha Shodhana procedures of Parada.	Practical Training	7	Shows how
Perform the Parada Jarana, Parada Murchana and Bandhana processes.	Practical Training	5	Shows how
Perform the Samanya Shodhana of Parada.	Experiential Learning	3	Shows how
DOES — 1			
Perform the Vishesha Shodhana procedures of Parada.	Experiential Learning	3	Does

Unit 2 · Parada Ashta Samskara and Ashtadasha Samskara

4 obj · 21 h · ceiling Shows how

2.1 Parada Samskara with research updates
2.2 Pharmaceutical equipment design for Parada Ashta Samskara
2.3 Hands-on experience of Parada Ashta Samskara

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe and discuss the utility of Ashtadasha Samskara.	Lecture	2	Knows how
Describe the design of pharmaceutical equipment to carry out Parada Ashta Samskara.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the Parada Samskara procedures- Swedana, Mardana, Murchana, and Utthapana.	Practical Training	8	Shows how
Elaborate and perform the Parada Ashta Samskara procedures- Patana, Bodhana Niyamana and Deepana.	Experiential Learning	10	Shows how

Unit 3 · Physico-chemical proprties and pharmacology of mercury and mercurial compounds

2 obj · 6 h · ceiling Shows how

3.1 Mercury metallurgy
3.2 Stoichiometry of mercury
3.3 Pharmacodynamics and pharmacokinetics of mercury and mercurial products

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe and elaborate on the metallurgy and stoichiometry of mercury.	Lecture	1	Knows
SHOWS HOW — 1			
Compare and analyze the Pharmacodynamics and pharmacokinetics of mercury and mercurial products.	Experiential Learning	5	Shows how

Unit 4 · Mercury toxicity, ADR and its management

3 obj · 7 h · ceiling Shows how

4.1 Mercury poisoning and its types

4.2 Ancient and Contemporary knowledge about the Treatment of Mercury poisoning

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the mercury poisoning and analyze its mechanism.	Lecture	1	Knows how
Describe the antidotes for mercury poisoning and its possible treatment by ancient and contemporary knowledge.	Lecture	1	Knows how
SHOWS HOW — 1			
Analyse mercury toxicity and ADR incidences.	Experiential Learning	5	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 4 · SEMESTER 4 · PAPER 1

Maharasa Dravya, its processing and utility

2 credits | 60 notional hours | 5 units | **22 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	12	9	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Explore and perform the pharmaceutical procedure, like Shodhana, Marana for Chapala/Kharpara, using authentic or substitute drugs.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
9 objectives

Analyze and Verify Maharasa Dravya Samples; Perform Shodhana, Marana, and other pharmaceutical procedures related to Abhraka and Vaikranta; Perform Shodhana, Marana, and other pharmaceutical procedures related to Makshika and Vimala; and 6 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
12 objectives

Describe and critically discuss the different Maharasa Dravyas mentioned in the classics; Analyze the technique of characterization for Maharasa Dravyas; Describe the significance of Abhraka Pancha Samskara; and 9 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Maharasa Dravyas with special reference to Abhraka and Vaikranta

8 obj · 18 h · ceiling Shows how

1.1 Comparative analysis of Maharasa Dravyas mentioned in various texts
1.2 Characterization of Maharasa dravyas
1.3 Abhraka
1.4 Vaikranta

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 6			
Describe and critically discuss the different Maharasa Dravyas mentioned in the classics.	Lecture	1	Knows how
Analyze the technique of characterization for Maharasa Dravyas.	Lecture	1	Knows how
Describe the significance of Abhraka Pancha Samskara.	Lecture	1	Knows how
Discuss the mineralogical identity of Vaikranta.	Lecture	1	Knows how
Analyze the pharmacotherapeutic attributes of Abhraka Bhasma and Vaikranta Bhasma.	Experiential Learning	5	Knows how
Analyse the pharmacology and clinical applications of Abhraka and Vaikranta.	Experiential Learning	2	Knows how
SHOWS HOW — 2			
Analyze and Verify Maharasa Dravya Samples.	Practical Training	2	Shows how
Perform Shodhana, Marana, and other pharmaceutical procedures related to Abhraka and Vaikranta.	Practical Training	5	Shows how

Unit 2 · Principles and Processing of Makshika and Vimala

3 obj · 11 h · ceiling Shows how

2.1 Critical understanding of Makshika (Copper Pyrite)
2.2 Detailed understanding of Vimala (Iron Pyrite)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe the therapeutic applications of Makshika and Vimala Bhasma, along with the antidotes for toxicity/ADR caused by them.	Lecture	1	Knows how
SHOWS HOW — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Makshika and Vimala.	Practical Training	5	Shows how
Perform the Shodhana of Makshika Bhasma and Vimala Bhasma.	Experiential Learning	5	Shows how

Unit 3 · Principles and Processing of Shilajit and Sasyaka

3 obj · 10 h · ceiling Shows how

3.1 Shilajatu (Black Bitumen)
3.2 Sasyaka(Bornite)/Tuttha(Blue Vitriol)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe and analyze the therapeutic applications of Shilajatu and Sasyaka/Tuttha, along with the antidotes for toxicity/ADR caused by them.	Lecture	1	Knows how
SHOWS HOW — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Shilajatu and Sasyaka/Tuttha.	Practical Training	3	Shows how
Perform the Shodhana of Shilajit and Sasyaka/Tuttha.	Experiential Learning	6	Shows how

Unit 4 · Principles and Processing of Chapala and Kharpara, along with physico-chemical analysis, with research updates of all Maharasa Dravyas

4 obj · 9 h · ceiling Does

4.2 Kharpara

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe the therapeutic applications of Chapala and Kharpara Bhasma, along with the antidotes for toxicity/ADR caused by them.	Lecture	1	Knows how
SHOWS HOW — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Chapala/ Kharpara.	Practical Training	2	Shows how
Demonstrate and perform physico-chemical analysis of Bhasma / Shodhita products of Maharasa Dravya category.	Practical Training	3	Shows how
DOES — 1			
Explore and perform the pharmaceutical procedure, like Shodhana, Marana for Chapala/ Kharpara, using authentic or substitute drugs.	Experiential Learning	3	Does

Unit 5 · Utility of new metals and minerals

4 obj · 8 h · ceiling Shows how

5.1 New metals/ minerals that align with Maharasa Dravyas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe the utility of new metals and minerals that align with Maharasa Dravyas.	Lecture	1	Knows how
Explore Shodhana process for the new metal/ minerals that align with Maharasa Dravyas.	Lecture	1	Knows how
Describe Marana process for the new metal/ minerals that align with Maharasa Dravyas.	Lecture	1	Knows how
SHOWS HOW — 1			
Identify and analyze New Metallic/Mineral Resources that align with Maharasa Dravyas Principles.	Experiential Learning	5	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **10 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 11 · SEMESTER 4 · PAPER 2

Mana Paribhasha (Metrology in Ayurvedic Pharmaceutical preparation)

2 credits | 60 notional hours | 5 units | **16 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1 certified here	5 certified here	7 certified here	3 certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	6	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Compare and judge the differences in the Mana mentioned by different classics; Analyze and justify the proportions of ingredients of classical formulations; Debate on merits and demerits of bulk production and the adaptation of advanced machines in Ayurvedic Pharmacies.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Apply the Knowledge of mana Paribhasha in the preparation of various formulations; Preparation of charts of unit converter (Ayurvedic Mana- Metric); Demonstrate preparation of classical formulations prepared with converted Units; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Interpret the application of Mana in classical formulations with a variety of examples; Describe standard units of weights and measures and AYUSH Unit Converter; Define the terms Volumetric methods and Gravimetric methods and apply the methods of both measurements to Ayurvedic Pharmaceutical Preparations; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Recall classical knowledge of Mana Paribhasha with a comparison of various classical texts. EC, LS, S.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Mana Paribhasha (Classical system of measurement)

4 obj · 12 h · ceiling Does

1.1 Importance of accurate measurements in proportions of ingredients

1.2 Importance of accurate measurements in dosage and administration of drugs

1.3 Types of Mana and details of classical units used for measurement (Weight/ Volume/ Length/ Time)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Recall classical knowledge of Mana Paribhasha with a comparison of various classical texts. EC, LS, S.	Lecture	1	Knows
KNOWS HOW — 1			
Interpret the application of Mana in classical formulations with a variety of examples.	Lecture	1	Knows how
SHOWS HOW — 1			
Apply the Knowledge of mana Paribhasha in the preparation of various formulations.	Practical Training	4	Shows how
DOES — 1			
Compare and judge the differences in the Mana mentioned by different classics.	Experiential Learning	6	Does

Unit 2 · Standard Units of Measurement

3 obj · 6 h · ceiling Shows how

2.1 AYUSH Unit Converter

2.2 Utility of converted units in the preparation of Classical formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe standard units of weights and measures and AYUSH Unit Converter.	Lecture	1	Knows how
SHOWS HOW — 2			
Preparation of charts of unit converter (Ayurvedic Mana- Metric).	Practical Training	2	Shows how
Demonstrate preparation of classical formulations prepared with converted Units.	Experiential Learning	3	Shows how

Unit 3 · Volumetric Vs Gravimetric Methods

3 obj · 6 h · ceiling Shows how

3.1 Key differences between Volumetric and gravimetric methods of measurement for a variety of drugs

3.2 Application of volumetric method for preparation of various classical formulations

3.3 Variation in proportion by using both methods and alteration in outcome

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Define the terms Volumetric methods and Gravimetric methods and apply the methods of both measurements to Ayurvedic Pharmaceutical Preparations.	Lecture	1	Knows how
SHOWS HOW — 2			
Apply and demonstrate the volumetric method for the preparation of classical formulations.	Practical Training	2	Shows how
Justify and interpret differences occurring in the final product prepared by varying the weight of the ingredients once taken by volume and once taken by mass(Gravimetric Method).	Experiential Learning	3	Shows how

Unit 4 · Rationality of Proportion of Ingredients in Ayurvedic formulations

3 obj · 16 h · ceiling Does

4.1 Factors accountable for deciding the proportion of ingredients of the

4.2 Profile of Raw Material (Fresh/ dry, Soft/ hard, part used, therapeutic properties, compatibility, etc.)

4.3 Rules for Duplication of ingredients in the same formulation

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe the classical rules and principles for the proportion of ingredients based on raw material and its nature/ Qualities.	Lecture	3	Knows how
SHOWS HOW — 1			
Demonstrate the skill of preparing formulations with a diverse range of raw materials.	Practical Training	6	Shows how
DOES — 1			
Analyze and justify the proportions of ingredients of classical formulations.	Experiential Learning	7	Does

Unit 5 · Bulk production of Ayurvedic Formulations

3 obj · 16 h · ceiling Does

5.1 Modifications expected for bulk production (in Material, Machine, Method, Mechanism)

5.2 Material- Type, proportion, effect of temperature pattern

5.3 Machine used for bulk manufacturing and automated drug manufacturing

5.4 Mechanism of machines used for bulk production and its merits and demerits

5.5 Methods used for bulk production in Ayurvedic Pharmacies and probable alterations in the qualities of the end products

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Enlist the machines, heating devices for bulk production. and describe their methods currently used by Ayurvedic pharmacies f.	Lecture	3	Knows how
SHOWS HOW — 1			
Demonstrate the use of machinery used for large-scale production in Ayurvedic pharmacies, with their mechanism.	Practical Training	6	Shows how
DOES — 1			
Debate on merits and demerits of bulk production and the adaptation of advanced machines in Ayurvedic Pharmacies.	Experiential Learning	7	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **10 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 12 · SEMESTER 4 · PAPER 2

Process validation Protocol

2 credits | 60 notional hours | 3 units | **10 objectives** | 50 formative marks | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	6	4	—
not used	certified here	certified here	not used

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	4	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	2	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
not used here

This module does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
4 objectives

Develop a process validation protocol for Ayurvedic dosage forms; Interpret the documentation related to Standard Operating Procedures (SOPs) used in pharmacies; Prepare different novel modern dosage forms of the given drug; and 1 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
6 objectives

Describe the SOP of Primary dosage forms, Secondary dosage forms, and topical dosage forms; Assess the significance of the Process Validation Protocol; Demonstrate the development of a process validation protocol for Ayurvedic dosage forms; and 3 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this module cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Pharmaceutical Preparation process: Classical Dosage forms

6 obj · 41 h · ceiling Shows how

1.1 Primary Dosage forms- Panchavidha Kashaya Kalpana and their Upakalpana

1.2 Secondary Dosage Forms- Avaleha Kalpana, Lavana Kalpana, Sneha Kalpana, Guti-vati-guggulu kalpana, Sandhana Kalpana, Arka-Kalpana, Kshara-kalpana

1.3 Topical Dosage forms- Lepa kalpana, Malahara Kalpana, Upanaha Kalpana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Describe the SOP of Primary dosage forms, Secondary dosage forms, and topical dosage forms.	Lecture	3	Knows how
Assess the significance of the Process Validation Protocol.	Lecture	2	Knows how
Demonstrate the development of a process validation protocol for Ayurvedic dosage forms.	Practical Training	10	Knows how
Prepare unique classical formulations following constructed SOP.	Experiential Learning	10	Knows how
SHOWS HOW — 2			
Develop a process validation protocol for Ayurvedic dosage forms.	Practical Training	10	Shows how
Interpret the documentation related to Standard Operating Procedures (SOPs) used in pharmacies.	Experiential Learning	6	Shows how

Unit 2 · Pharmaceutical Preparation process: Recent advances of Dosage forms

3 obj · 10 h · ceiling Shows how

2.1 Principles of processes viz. Expression, Extraction, Maceration,

2.2 Solid Dosage forms- Powders, Pills, Tablets, Capsules, Soft gel capsules, Gummies, Lozenges, Granules

2.3 Liquid Dosage Forms- Emulsions, Suspensions, Lotions, Liniments, Elixir, Syrup, Nasal Drops, Gargle

2.4 Semi-solid dosage forms- Ointment, Gel, Paste, Cream, Plasters, Suppositories

2.5 Gaseous dosage forms- Aerosols, Nasal Spray, Inhalers

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe principles of different dosage forms, viz. Extraction, Maceration, Percolation Infusion, Distillation, Active principle isolation, Particle size reduction, Separation techniques, and Mixing techniques.	Lecture	2	Knows how
Discuss Pharmaceutical Preparation Process of Solid, semisolid, liquid, and gaseous dosage forms.	Lecture	3	Knows how
SHOWS HOW — 1			
Prepare different novel modern dosage forms of the given drug.	Experiential Learning	5	Shows how

Unit 3 · Novel Dosage and drug delivery Forms

1 obj · 5 h · ceiling Shows how

3.1 Importance and Need of Different Dosage Forms in Drug Delivery Systems and Treatment Modalities

3.2 Definition, significance, technologies driving Novel dosage forms, applications, and challenges

3.3 Modification of different dosage forms

3.4 Potential areas for future research and development

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
SHOWS HOW — 1			
Analyze, justify, and demonstrate the importance and need of the preparation of Novel dosage forms.	Experiential Learning	5	Shows how

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **4 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Shows how.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 19 · SEMESTER 4 · PAPER 3

SOP and clinical application of Chaturvidha Rasayana

3 credits | 90 notional hours | 4 units | **25 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	13	9	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	14	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	7	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Evaulate the pharmacodynamic rationale and clinical application of parpati kalpana; Prepare Pottali following SOP.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
9 objectives

Demonstrate the method of preparation and SOP of Kharaleeya Rasayana; Appraise the utility of kharaleeya rasayana to clinical case practice; Demonstrate the impact of Bhavana (A bioactivation tool) in Kharaleeya Rasayana; and 6 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
13 objectives

Describe the unique features, types, and significance of parpati kalpana; Discuss the role of bhavana (pharmacodynamics of trituration) in potentiating kharaleeya rasayana; Describe the unique features, types, and significance of parpati kalpana; and 10 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Review and analyze research updates on Parpati Kalpana.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Standard Operating Procedure (SOP) and Clinical Application of Khalveeya Rasayana

5 obj · 17 h · ceiling Shows how

1.1 Definition and significance of Khalveeya Rasayana
1.2 Standard Operating Procedure for the preparation of Khalveeya Rasayana
1.3 Role of bhavana in potentiating Khalveeya rasayana
1.4 Clinical application of Khalveeya Rasayana with examples

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the unique features, types, and significance of parpati kalpana.	Lecture	2	Knows how
Discuss the role of bhavana (pharmacodynamics of trituration) in potentiating kharaleeya rasayana.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the method of preparation and SOP of Kharaleeya Rasayana.	Practical Training	7	Shows how
Appraise the utility of kharaleeya rasayana to clinical case practice.	Experiential Learning	4	Shows how
Demonstrate the impact of Bhavana (A bioactivation tool) in Kharaleeya Rasayana.	Experiential Learning	3	Shows how

Unit 2 · SOP and clinical application of Parpati Rasayana

6 obj · 15 h · ceiling Does

2.1 Definition and significance of Parpati
2.2 Standard Operating Procedure for the preparation of Parpati with process validation
2.3 Mechanism of action of Parpati kalpana
2.4 Pathyapathya in Parpati Prayoga and its significance
2.5 Clinical application of Parpati Kalpa with examples

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Review and analyze research updates on Parpati Kalpana.	Lecture	1	Knows
KNOWS HOW — 3			
Describe the unique features, types, and significance of parpati kalpana.	Lecture	1	Knows how
Discuss SOP and quality control parameters of Parpati Kalpa.	Lecture	1	Knows how
Discuss in detail the pathya-apathya, matra, prayoga of parpati kalpana.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the SOP of parpati kalpana.	Practical Training	7	Shows how
DOES — 1			
Evaulate the pharmacodynamic rationale and clinical application of parpati kalpana.	Experiential Learning	4	Does

Unit 3 · SOP and clinical application of Pottali Rasayana

7 obj · 25 h · ceiling Does

3.1 Definition and significance of Pottali rasayana
3.2 Standard Operating Procedure for the preparation of Pottali Kalpa with process validation
3.3 Role of Gandhaka drava paka in the preparation Pottali rasayana
3.4 Special clinical attributes of Pottali with examples

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Define pottali and discuss in detail the concept, types, and significance of pottali rasayana.	Lecture	1	Knows how
Discuss in detail the various classical methods of preparation of pottali kalpana.	Lecture	1	Knows how
Discuss the standardization and quality control parameters for Pottali Kalpa.	Lecture	1	Knows how
Evalaute the safety and efficacy of Pottali Rasayana.	Lecture	1	Knows how

SHOWS HOW — 2			
Demonstrate the method of preparation and SOP of pottali kalpana.	Practical Training	8	Shows how
Analyse the pharmacological profile of the ingredients, bhavana dravya, and the pharmaceutical procedure adopted in achieving therapeutic precision.	Experiential Learning	5	Shows how
DOES — 1			
Prepare Pottali following SOP.	Experiential Learning	8	Does

Unit 4 · SOP and clinical application of Kupipakwa Rasayana

7 obj · 27 h · ceiling Shows how

4.1 Definition and significance of Kupipakwa rasayana
4.2 Standard Operative Procedure for the preparation of Kupipakwa Rasayana with process validation
4.3 Role of Kramagni paka in preparation of Kupipakwa Rasayana
4.4 Chemistry of Kupipakwa rasayana
4.5 Clinical application of Kupipakwa Rasayana Kalpa with examples

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Discuss the concept and significance of Kupipakwa Rasayana.	Lecture	1	Knows how
Describe in detail the role of Kramagni Paka in the preparation of Kupipakwa Rasayana.	Lecture	1	Knows how
Describe the role of gandhaka jarana in the context of sagandha kupipakwa rasayana.	Lecture	1	Knows how
Discuss the thermo–chemistry of kupipakwa rasayana.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the method of preparation and SOP of Kupipakwa Rasayana.	Practical Training	8	Shows how
Prepare the selected Kupipakwa Rasayana.	Experiential Learning	10	Shows how
Demonstrate the scientific rationality for the selection of specific Kupipakwa Rasayana in the given scenario.	Experiential Learning	5	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **11 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 20 · SEMESTER 4 · PAPER 3

SOP and Clinical application of Ayaskriti, Louha, and Mandura Kalpa

1 credits | 30 notional hours | 3 units | **12 objectives** | 25 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	6	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Evaluate the role of Gomutra in pharmaceutical processing and as a bio enhancer.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate the SOP and preparation of Ayaskriti; Evaluate the role of Ayaskriti.as a multi-target formulation; Demonstrate the SOP and preparation of Louha Kalpa; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Discuss in detail the concept of Ayaskriti, its types, method of administration, and safety concerns; Discuss the concept of Louha Kalpa; Discuss the key chemical considerations in Louha Kalpa; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · SOP and clinical application of Ayaskriti

3 obj · 8 h · ceiling Shows how

1.1 Ayaskriti
1.2 Standard Operating Procedure for the preparation of Ayaskriti
1.3 Types (forms) of ayaskriti
1.4 Method of administration and clinical utility of Ayaskriti
1.5 Safety concerns of Ayaskriti

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss in detail the concept of Ayaskriti, its types, method of administration, and safety concerns.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the SOP and preparation of Ayaskriti.	Practical Training	3	Shows how
Evaluate the role of Ayaskriti.as a multi-target formulation.	Experiential Learning	4	Shows how

Unit 2 · SOP and clinical application of Louha kalpa

4 obj · 9 h · ceiling Shows how

2.1 Pharmaceutical vistas of Louha Kalpas
2.2 Key chemical considerations in Louha Kalpa
2.3 Pharmacology and clinical utility of Louha kalpa

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the concept of Louha Kalpa.	Lecture	1	Knows how
Discuss the key chemical considerations in Louha Kalpa.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the SOP and preparation of Louha Kalpa.	Practical Training	3	Shows how
Apply Louha Kalpa by integrating through textual insights with real word cases for the therapeutic purpose.	Experiential Learning	4	Shows how

Unit 3 · SOP and Clinical application of Mandura Kalpa

5 obj · 11 h · ceiling Does

3.1 Concept of mandura kalpa and its SOP
3.2 Types of mandura kalpa based on the method of preparation
3.3 Significance of go mutra in mandura kalpa
3.4 Clinical application of mandura kalpa

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the concept and pharmaceutical vistas of Mandura Kalpa.	Lecture	1	Knows how
Describe chemistry and pharmacology associated with structure and preparation of Mandura Kalpas.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the SOP and preparation of Mandura Kalpa.	Practical Training	2	Shows how
Demonstrate the clinical application of Mandura Kalpa.	Practical Training	2	Shows how
DOES — 1			
Evaluate the role of Gomutra in pharmaceutical processing and as a bio enhancer.	Experiential Learning	5	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **7 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 27 · SEMESTER 4 · PAPER 4

Network pharmacology & Reverse Pharmacology

2 credits | 60 notional hours | 2 units | **16 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	4	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	2	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Practice Network Pharmacology; Carry out a mini-project on network pharmacology.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
4 objectives

Demonstrate network pharmacology applications with the help of a video; Demonstrate reverse pharmacology applications with the help of a video; Present observations of analysis of Reverse Pharmacology in selected cases; and 1 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Define network pharmacology and differentiate it from traditional pharmacology; Describe network construction from herbal compounds; Discuss and analyze Research updates on Reverse pharmacology; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Recognize key network concepts like nodes (genes/proteins/drugs) and edges(interactions).

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Netwok Pharmacology

8 obj · 28 h · ceiling Does

1.1 Fundamental knowledge of Network pharmacology
1.2 Network pharmacology concepts
1.3 Computational methods
1.4 Applications
1.5 Research updates on Network pharmacology

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Recognize key network concepts like nodes (genes/proteins/drugs) and edges(interactions).	Lecture	1	Knows
KNOWS HOW — 4			
Define network pharmacology and differentiate it from traditional pharmacology.	Lecture	1	Knows how
Describe network construction from herbal compounds.	Lecture	1	Knows how
Discuss and analyze Research updates on Reverse pharmacology.	Lecture	1	Knows how
Discuss various applications and research updates on Network pharmacology.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate network pharmacology applications with the help of a video.	Practical Training	10	Shows how
DOES — 2			
Practice Network Pharmacology.	Experiential Learning	10	Does
Carry out a mini-project on network pharmacology.	Experiential Learning	3	Does

Unit 2 · Reverse Pharmacology

8 obj · 28 h · ceiling Shows how

2.1 Introduction, Definition of Reverse Pharmacology, History, and Scope
2.2 Drug Discovery Current Scenario, Challenges, and Future Perspective
2.3 Reverse pharmacology for herbal drug-based modern medicines
2.4 Research updates on Reverse pharmacology

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe the definition and history of reverse pharmacology and discuss its Scope.	Lecture	1	Knows how
Elaborate current scenario, challenges, and future perspectives of reverse pharmacology.	Lecture	1	Knows how
Assess reverse pharmacology leverages traditional medicine and clinical experience to accelerate drug development.	Lecture	1	Knows how
Apply Reverse pharmacology for herbal-based modern medicines.	Lecture	1	Knows how
Discuss and analyze Research updates on Reverse pharmacology.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate reverse pharmacology applications with the help of a video.	Practical Training	10	Shows how
Present observations of analysis of Reverse Pharmacology in selected cases.	Experiential Learning	10	Shows how
Validate Reverse pharmacology with a systems biology approach.	Experiential Learning	3	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **6 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 28 · SEMESTER 4 · PAPER 4

Pharmaceutical Quality Assurance

2 credits | 60 notional hours | 5 units | **19 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	6	3
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Inspect Packaging Materials to check their compliance with quality specifications; Perform calibration of Analytical Instruments; Conduct QA Documentation audit.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate the process of mock GMP Inspection; Demonstrate GMP Hierarchy and Responsibilities; Identify the Root cause, affected personnel roles, and suggest corrective and preventive measures through Case Study Discussion on GMP Violation; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Define and elaborate the concept of Quality Assurance and Quality Management as per; Describe Total Quality Management (TQM) and ICH Guidelines, QSEM, and ISO 9000; Illustrate the Steps for registration of NABL accreditation; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Elaborate GLP with its Principles and Requirements, documentation, Regulatory compliance, Audit procedures.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Quality Assurance and Quality Management concepts

6 obj · 19 h · ceiling Shows how

1.1 Introduction to Quality Assurance and Quality Management concepts
1.2 Definition and concept of Quality control,
1.3 Quality assurance and GMP
1.4 Total Quality Management (TQM)
1.5 ICH Guidelines
1.6 Brief overview of Pharmaceutical industry guidelines for Quality, Safety, Efficacy and Multidisciplinary (QSEM)

1.7 Overview of ISO 9000 & ISO14000

1.8 Steps for registration of NABL accreditation

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Define and elaborate the concept of Quality Assurance and Quality Management as per.	Lecture	1	Knows how
Describe Total Quality Management (TQM) and ICH Guidelines, QSEM, and ISO 9000.	Lecture	1	Knows how
Illustrate the Steps for registration of NABL accreditation.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the process of mock GMP Inspection.	Practical Training	5	Shows how
Demonstrate GMP Hierarchy and Responsibilities.	Practical Training	5	Shows how
Identify the Root cause, affected personnel roles, and suggest corrective and preventive measures through Case Study Discussion on GMP Violation.	Experiential Learning	6	Shows how

Unit 2 · Personnel Responsibilities

3 obj · 8 h · ceiling Shows how

2.1 Organization and personnel

2.2 Personnel responsibilities, training, hygiene, and personal records

2.3 Premises: Design, construction, and plant layout, maintenance, sanitation, environmental control, utilities, and maintenance of sterile areas, control of contamination

2.4 Equipment and raw materials: Equipment selection, purchase specifications, maintenance, and purchase specifications and maintenance of stores for raw materials

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe Organizational structure and personnel management.	Lecture	1	Knows how
Discuss design, construction, and maintenance of Premises.	Lecture	1	Knows how
SHOWS HOW — 1			
Document a personal record of pharmacy personnel.	Experiential Learning	6	Shows how

Unit 3 · Containers, Packing materials

4 obj · 13 h · ceiling Does

3.1 Types of Packaging

3.2 Packaging materials

3.3 Container types and closures

3.4 Packaging and stability

3.5 Regulatory aspects

3.6 Quality Control

3.7 Smart Packaging

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss selection and maintenance of equipment, purchase specifications of raw materials, and maintenance of stores.	Lecture	1	Knows how
Discuss Types of containers and packaging, Regulatory aspects, their Quality Control and Smart Packaging.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the identification and Classification of packaging materials.	Practical Training	5	Shows how
DOES — 1			
Inspect Packaging Materials to check their compliance with quality specifications.	Experiential Learning	6	Does

Unit 4 · Good Laboratory Practices

3 obj · 8 h · ceiling Does

4.1 GLP Principles and Requirements

4.2 GLP documentation

4.3 Regulatory compliance

4.4 Audit procedures

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Elaborate GLP with its Principles and Requirements, documentation, Regulatory compliance, Audit procedures.	Lecture	1	Knows
SHOWS HOW — 1			
Demonstrate SOP Writing for different equipment.	Practical Training	5	Shows how
DOES — 1			
Perform calibration of Analytical Instruments.	Experiential Learning	2	Does

Unit 5 · Document maintenance in pharmaceutical industry

3 obj · 8 h · ceiling Does

4.1 Documentation related to Grievances and Redressal

4.2 Complaints and evaluation of complaints, Handling of returned goods, recalling, and waste disposal

4.3 Document maintenance in the pharmaceutical industry: Batch Formula Record, Master Formula Record, SOP, Quality audit, Quality Review, Quality documentation,

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Define Complaints, their evaluation, product recall, and waste disposal.	Lecture	1	Knows how
Describe document maintenance in the pharmaceutical industry.	Lecture	1	Knows how
DOES — 1			
Conduct QA Documentation audit.	Experiential Learning	6	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **9 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 5 · SEMESTER 5 · PAPER 1

Critical appraisal of Uparasa Dravya and Sadharana Rasa Dravya

2 credits | 60 notional hours | 2 units | **24 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	10	12	2
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	8	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Prepare and present the formulations of Sadharana Rasa Dravya; Assemble the utility of Sadharana Dravyas in the clinical practice through a Field Survey of manufacturers and clinicians.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
12 objectives

Demonstrate the authentication of Uparasa Dravyas; Demonstrate Shodhana, Marana and other pharmaceutical procedures for Uparasa; Demonstrate Quality Control testing for the products of Uparasa Dravya; and 9 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
10 objectives

Describe he different Uparasa Dravyas mentioned in the classics; Discuss the controversial drugs under Uparasa category; Discuss the mineralogical characterization of Uparasa; and 7 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Principle and Processing of Uparasa Dravya

12 obj · 28 h · ceiling Shows how

1.1 Uparasa Dravyas mentioned in various texts
1.2 Characterization of Uparasa Dravyas
1.3 Pharmaceutical processes of Uparasa Dravya
1.4 Pharmacology and clinical application of Uparasa Dravya

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe he different Uparasa Dravyas mentioned in the classics.	Lecture	1	Knows how
Discuss the controversial drugs under Uparasa category.	Lecture	1	Knows how
Discuss the mineralogical characterization of Uparasa.	Lecture	1	Knows how
Recognise the therapeutic applications and toxico-kinetics of Uparasa Dravyas.	Lecture	1	Knows how
Discuss research updates on Uparasa drugs.	Lecture	1	Knows how
SHOWS HOW — 7			
Demonstrate the authentication of Uparasa Dravyas.	Practical Training	2	Shows how
Demonstrate Shodhana, Marana and other pharmaceutical procedures for Uparasa.	Practical Training	4	Shows how
Demonstrate Quality Control testing for the products of Uparasa Dravya.	Practical Training	4	Shows how
Identify and authenticate Uparasa Dravyas.	Experiential Learning	3	Shows how
Perform Quality Control tests for the products of Uparasa Dravya.	Experiential Learning	3	Shows how
Prepare formulations of Uparasa Dravya.	Experiential Learning	5	Shows how
Review articles on the toxicity and ADRs of Haratala and Manahshila and interpret.	Experiential Learning	2	Shows how

Unit 2 · Principle and Processing of Sadharana Rasa

12 obj · 28 h · ceiling Does

2.1 Sadharana rasa Dravyas mentioned in various texts
2.2 Characterization of Sadharana Rasa Dravyas
2.3 Pharmaceutical processes of Sadharana rasa Dravya
2.4 Pharmacology and clinical application of Sadharana rasa Dravya

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe and critically discuss the different Sadharana Rasa Dravyas mentioned in the classics.	Lecture	1	Knows how
Analyze the controversial drugs under Sadharana Rasa category.	Lecture	1	Knows how
Discuss the mineralogical characterization of Sadharana Rasa.	Lecture	1	Knows how
Recognise the therapeutic applications and toxico-kinetics of Sadharana Rasa Dravyas.	Lecture	1	Knows how
Discuss research updates on Sadharana Rasa drugs.	Lecture	1	Knows how

SHOWS HOW — 5			
Demonstrate authentication of Sadharana Rasa Dravya.	Practical Training	2	Shows how
Demonstratie Shodhana, Marana and other pharmaceutical procedures for Sadharana.	Practical Training	4	Shows how
Demonstrate Quality Control testing for the products of Sadharana Rasa Dravya.	Practical Training	4	Shows how
Identify and authenticate Sadharana Rasa Dravyas.	Experiential Learning	3	Shows how
Perform Quality Control tests for the products of Sadharana Rasa Dravya.	Experiential Learning	3	Shows how
DOES — 2			
Prepare and present the formulations of Sadharana Rasa Dravya.	Experiential Learning	3	Does
Assemble the utility of Sadharana Dravyas in the clinical practice through a Field Survey of manufacturers and clinicians.	Experiential Learning	4	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **14 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 6 · SEMESTER 5 · PAPER 1

Critical appraisal of Dhatu and Updhatu

2 credits | 60 notional hours | 2 units | **24 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	10	11	3
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	8	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Identify and document classical Dhatus; Prepare formuations using Dhatu Bhasma; Prepare Upadhatu formulations.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
11 objectives

Demonstrate the authentication of Dhatu Dravya; Perform pharmaceutical processing (Shodhana & Marana) of metals; Demonstrate Quality control of metallic preparations; and 8 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
10 objectives

Describe classical Dhatu Dravyas; Discuss the controversial metals in Ayurveda texts; Characterize Dhatu Dravyas mineralogically and pharmaceutically; and 7 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Principle and Processing of Dhatu / Metals

12 obj · 28 h · ceiling Does

1.1 Dhatus mentioned in Rasa classics
1.2 Classification and characterization of metals
1.3 Pharmaceutical processing (Shodhana, Marana, etc.)
1.4 Therapeutic uses, toxicity, and safety of metallic preparations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe classical Dhatu Dravyas.	Lecture	1	Knows how
Discuss the controversial metals in Ayurveda texts.	Lecture	1	Knows how
Characterize Dhatu Dravyas mineralogically and pharmaceutically.	Lecture	1	Knows how
Identify therapeutic applications and toxico-kinetics of Dhatu.	Lecture	1	Knows how
Compile and interpret research and regulatory updates on metal-based Ayurvedic drugs.	Lecture	1	Knows how
SHOWS HOW — 5			
Demonstrate the authentication of Dhatu Dravya.	Practical Training	2	Shows how
Perform pharmaceutical processing (Shodhana & Marana) of metals.	Practical Training	4	Shows how
Demonstrate Quality control of metallic preparations.	Practical Training	4	Shows how
Perform Quality assessment of Dhatu Bhasma.	Experiential Learning	3	Shows how
Identify and analyze Adverse drug reactions (ADRs) due to metals.	Experiential Learning	2	Shows how
DOES — 2			
Identify and document classical Dhatus.	Experiential Learning	3	Does
Prepare formuations using Dhatu Bhasma.	Experiential Learning	5	Does

Unit 2 · Principle and Processing of Upadhatu

12 obj · 28 h · ceiling Does

2.1 Classical analysis of Upadhatu
2.2 Classification, sources, and characteristics of Upadhatu
2.3 Pharmaceutical processing (Shodhana, Marana, etc.)of Upadhatu
2.4 Pharmacological actions and clinical application of Upadhatu

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe and classify Upadhatu from Rasashastra references.	Lecture	1	Knows how
Discuss controversial and newly interpreted Upadhatu.	Lecture	1	Knows how
Analyze Upadhatu mineralogically and pharmaceutically.	Lecture	1	Knows how
Recognize therapeutic applications and possible ADRs of Upadhatu.	Lecture	1	Knows how
Review and interpret findings of research and current practices involving Upadhatu.	Lecture	1	Knows how

SHOWS HOW — 6			
Demonstrate the Identification of Upadhatu samples.	Practical Training	2	Shows how
Demonstrate Shodhana and Marana of Upadhatu.	Practical Training	4	Shows how
Demonstrate the Quality control tests for sub-metallic formulations.	Practical Training	4	Shows how
Identify the Upadhatu samples.	Experiential Learning	3	Shows how
Perform Quality Control tests for the products of Upadhatu.	Experiential Learning	3	Shows how
Analyze and interpret findings of the utility of Dhatu and Upadhatu in clinical practice through a Field survey with Manufacturers or Vaidyas.	Experiential Learning	4	Shows how
DOES — 1			
Prepare Upadhatu formulations.	Experiential Learning	3	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **14 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 13 · SEMESTER 5 · PAPER 2

Krutanna (Ayurvedic Food Preparations)

2 credits | 60 notional hours | 4 units | **15 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	4	6	4
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
4 objectives

Prepare distinctive formulations using Aahara Dravya from different groups (Vargas); Develop the disease-specific Personalised prescription of Aushadhi Siddha Ahara; Demonstrate the preparation of novel Ayurvedic food products; and 1 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate the method of preparation of distinctive formulations using Aahara Dravya from different groups (Vargas); Demonstrate preparation of therapeutic diet formulations; Identify and prepare disease specific Ayurvedic Ahara; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Describe Ayurvedic dietary principles and the significance of food substances used as medicine; Describe disease-specific diet formulations; Recite paribhasha and describe the importance of Yusha, Yavagu, Odana, and Mamsarasa; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Describe the regulation standards for Ayurvedic food preparations- FSSAI (Ayurveda Aahara regulations).

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Classification of food substances and Formulations

3 obj · 12 h · ceiling Does

1.1 Ayurvedic dietary principles and Ahara Dravya Varga (Groups of food substance)

1.2 Aahara -mahabhaishajya: Significance of food substances used as medicine

1.3 Unique classical formulations prepared using Food substances belonging to various groups and their utility in treatment

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe Ayurvedic dietary principles and the significance of food substances used as medicine.	Lecture	2	Knows how
SHOWS HOW — 1			
Demonstrate the method of preparation of distinctive formulations using Aahara Dravya from different groups (Vargas).	Practical Training	4	Shows how
DOES — 1			
Prepare distinctive formulations using Aahara Dravya from different groups (Vargas).	Experiential Learning	6	Does

Unit 2 · Customized food for treatment

6 obj · 24 h · ceiling Does

2.1 Disease-specific Ayurvedic Aahara- Therapeutic utility

2.2 Samsarjana krama- Manda, Peya, Vilepi, Yavagu, Yusha, Odana, Krushara, Mansarasa- Process validation Protocol for all formulations

2.3 Personalised prescription of Aushada Siddha Aahara for Shamana and during panchakarma

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe disease-specific diet formulations.	Lecture	2	Knows how
Recite paribhasha and describe the importance of Yusha, Yavagu, Odana, and Mamsarasa.	Lecture	2	Knows how
SHOWS HOW — 3			
Demonstrate preparation of therapeutic diet formulations.	Practical Training	4	Shows how
Identify and prepare disease specific Ayurvedic Ahara.	Experiential Learning	6	Shows how
Demonstrate the method of preparation of various Aahara Kalpana.	Practical Training	4	Shows how
DOES — 1			
Develop the disease-specific Personalised prescription of Aushadhi Siddha Ahara.	Experiential Learning	6	Does

Unit 3 · Ayurvedic Food Industry

3 obj · 10 h · ceiling Does

3.1 Market survey about Ayurvedic food products and research updates

3.2 Global needs and designing of modified food formulations

3.3 Semi-processed food/ semi-cooked food, RTC- ready to cook / RTD-Ready to drink Foods and drinks

3.4 Regulation Standards for Ayurvedic Food Preparation (Ayurveda Aahara Regulations- 2021), FSSAI Regulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the regulation standards for Ayurvedic food preparations- FSSAI (Ayurveda Aahara regulations).	Lecture	2	Knows
DOES — 2			
Demonstrate the preparation of novel Ayurvedic food products.	Practical Training	4	Does
Collect the data and interpret the findings through a market survey to explore the available Ayurvedic food products and assess the current trends in the Ayurvedic food industry.	Experiential Learning	4	Does

Unit 4 · Food processing technology traditional and modern methods

3 obj · 10 h · ceiling Shows how

4.1 Traditional Methods- Steaming, Boiling, Drying, Salting, Smoking, Frying,

4.2 Modern Methods- Heating, Freeze Drying, Vacuum Drying, Canning, Juicing, Fermentation, Modified atmosphere packaging, Pasteurisation, Smoking, Pulse electric

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Differentiate between classical and contemporary food processing techniques and support the current and emerging trends in the Ayurvedic food industry.	Lecture	2	Knows how
SHOWS HOW — 2			
Demonstrate various food processing techniques used in current Ayurvedic food Industries.	Practical Training	4	Shows how
Analyze Functional Food and Modified Food Formulations in Ayurvedic Food Industry.	Experiential Learning	4	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **10 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 14 · SEMESTER 5 · PAPER 2

Ayurvedic Formulations for Holistic Wellbeing, Vitality and Body Care

2 credits | 60 notional hours | 2 units | **11 objectives** | 50 formative marks | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	4	7	—
not used	certified here	certified here	not used

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	4	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	3	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
not used here

This module does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate the preparation method of Rasayana Drugs following validated SOP; Demonstrate the role of Ayurvedic Rasayana formulations in the maintenance of health and Vajikarana Formulations for better progeny(Suprajanana); Select appropriate Ayurvedic Formulations for the fitness of specific body parts and analyze their role in the maintenance of health; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Describe the mode of action of Rasayana Formulations; Describe classical Rasayana and Vajikarana Formulations; Discuss the role of Ayurvedic formulations in Cosmetic preparations; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this module cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Rasayana-Personalized Rejuvenation Formulations

5 obj · 24 h · ceiling Shows how

1.1 Key aspects of Rasayana Formulations

1.2 General Rasayana and Organ-specific/Target-specific Rasayana- Varnya, Tvachya, Kanthya, Swarya, Keshya, Hrudya, Medhya- Formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the mode of action of Rasayana Formulations.	Lecture	2	Knows how
Describe classical Rasayana and Vajikarana Formulations.	Lecture	2	Knows how
SHOWS HOW — 3			
Demonstrate the preparation method of Rasayana Drugs following validated SOP.	Practical Training	8	Shows how
Demonstrate the role of Ayurvedic Rasayana formulations in the maintenance of health and Vajikarana Formulations for better progeny(Suprajanana).	Experiential Learning	5	Shows how
Select appropriate Ayurvedic Formulations for the fitness of specific body parts and analyze their role in the maintenance of health.	Experiential Learning	7	Shows how

Unit 2 · Ayurvedic based cosmetic preparation

6 obj · 32 h · ceiling Shows how

2.1 Body/ Hair/ Lip/ Oral/Skin/ Foot care products- Preparation methods and application in Ayurvedic

2.2 Preparation process of Lip balm, lipsticks, shampoos, cold cream and vanishing cream, toothpaste, hair dyes, Sunscreen lotion, Face pack, Mask, Toners, herbal soaps,

2.3 Current and Emerging trends – Ayurvastram, Ayurvedic footwear, Eye mask

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the role of Ayurvedic formulations in Cosmetic preparations.	Lecture	3	Knows how
Evaluate the Global need for body care products in the healthcare sector.	Lecture	3	Knows how
SHOWS HOW — 4			
Demonstrate preparation of Herbal Cosmetics.	Practical Training	6	Shows how
Demonstrate the preparation of Modified Cosmeceuticals.	Practical Training	6	Shows how
Analyze different cosmetics available in the market.	Experiential Learning	7	Shows how
Design and prepare novel Ayurvedic cosmeceuticals.	Experiential Learning	7	Shows how

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **7 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Shows how.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 21 · SEMESTER 5 · PAPER 3

Clinical application of Rasoushadhi according to Srotas and in selected challenging clinical conditions

1 credits | 30 notional hours | 3 units | **13 objectives** | 25 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	7	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Explore the role of Rasoushadhi in clinically challenging conditions such as neurodevelopmental and neurodegenerative diseases.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate the clinical application of Rasoushadhi in selected Srotas, illustrating their practical utility and therapeutic outcomes in addressing specific health conditions; Illustrate the therapeutic application of Rasoushadhi in different Srotas; Demonstrate the clinical applicability of Rasoushadhi in Medical Emergencies; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Discuss Srotas-specific Rasoushadhi, tailored Ayurvedic formulations for individual biological channels; Enlist and discuss Rasoushadhi for different Srotas, explaining their therapeutic efficacy and mechanism of action; Discuss the unique properties of Rasoushadhi, enabling its use in clinical emergencies; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Clinical application of Rasoushadhis specific to Srotas

4 obj · 8 h · ceiling Shows how

1.1 Concept and rationality of selection and application of Rasoushadhi according to Srotas

1.2 Clinical application of Rasouashdhi in disorders of different Srotas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss Srotas-specific Rasoushadhi, tailored Ayurvedic formulations for individual biological channels.	Lecture	1	Knows how
Enlist and discuss Rasoushadhi for different Srotas, explaining their therapeutic efficacy and mechanism of action.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the clinical application of Rasoushadhi in selected Srotas, illustrating their practical utility and therapeutic outcomes in addressing specific health conditions.	Practical Training	3	Shows how
Illustrate the therapeutic application of Rasoushadhi in different Srotas.	Experiential Learning	3	Shows how

Unit 2 · Clinical application of Rasoushadhi in emergency conditions

4 obj · 9 h · ceiling Shows how

2.1 Role of Rasoushadhis in Managing Medical Emergencies

2.2 Specific Rasa Yogas among Chaturvidha Rasayana kalpas for Managing Medical Emergencies

2.3 Application of Pottali Kalpa in Emergency Management

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the unique properties of Rasoushadhi, enabling its use in clinical emergencies.	Lecture	1	Knows how
Enlist specific Rasa Yogas for managing medical emergencies and discuss their rationality, highlighting the therapeutic mechanisms and clinical applications.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the clinical applicability of Rasoushadhi in Medical Emergencies.	Practical Training	3	Shows how
Apply the appropriate Rasa Oushadhi in a alloted /selected Medical Emergency conditions.	Experiential Learning	4	Shows how

Unit 3 · Rasoushadhi in specific, clinically challenging conditions

5 obj · 11 h · ceiling Does

3.1 Rational application of Rasoushadhi in cancer

3.2 Application of Rasoushadhi in neurodevelopmental conditions

3.3 Utility of Rasoushadhi in neurodegenerative conditions

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss the role of Rasoushadhi in clinically challenging conditions like cancers neurodevelopmental and neurodegenerative disorders.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the role of Rasoushadhi in clinically challenging conditions.	Practical Training	2	Shows how
Demonstrate the role of Rasoushadhi as Rasayana in clinically challenging conditions.	Practical Training	2	Shows how
Evalaute the role of Rasoushadhi in clinically challenging conditions such as Cancer.	Experiential Learning	3	Shows how
DOES — 1			
Explore the role of Rasoushadhi in clinically challenging conditions such as neurodevelopmental and neurodegenerative diseases.	Experiential Learning	3	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 22 · SEMESTER 5 · PAPER 3

Yogas from different Kalpanas

3 credits | 90 notional hours | 5 units | **27 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	16	8	3
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	15	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Analyse the role of selected leha yoga in different clinical conditions; Illusrate the clinical application of selected Arka/ Dravaka Kalpa; Perform Comparative analysis of Asava Arishta Samples having different percentage and types of sugar.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
8 objectives

Demonstrate the preparation of Vati, Guggulu Kalpa, and their modified modern dosage forms; Design an S.O.P for the preparation of Ayurvedic suppository specifying its clinical utility based on Varti Kalpana; Demonstrate the role of pharmaceutical additives and Samyak Paka of Avaleha; and 5 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
16 objectives

Describe the therapeutic significance of vati kalpana; Summarize the therapeutic significance of Guggulu Kalpa and Varti Kalpana; Discuss modern aspects of solid dosage forms; and 13 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · SOP of Vati, Gutika, Guggulu and Varti Kalpas and their therapeutic applications

6 obj · 20 h · ceiling Shows how

1.1 SOP of selected Vati preparation including packing,
1.2 SOP regarding preparation of selected formulations from Guggulu Kalpana, including packing, storage, and therapeutic applications
1.3 Types of Varti and their SOP
1.4 Clinical applications of different Varti
1.5 Suppositories and pharmaceutical modifications of Ayurvedic suppositories

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Describe the therapeutic significance of vati kalpana.	Lecture	1	Knows how
Summarize the therapeutic significance of Guggulu Kalpa and Varti Kalpana.	Lecture	1	Knows how
Discuss modern aspects of solid dosage forms.	Lecture	1	Knows how
Analyse the role of Gugulu Kalpana in various clinical conditions by selecting a specific Gulggulu Yoga.	Experiential Learning	8	Knows how
SHOWS HOW — 2			
Demonstrate the preparation of Vati, Guggulu Kalpa, and their modified modern dosage forms.	Practical Training	5	Shows how
Design an S.O.P for the preparation of Ayurvedic suppository specifying its clinical utility based on Varti Kalpana.	Practical Training	4	Shows how

Unit 2 · SOP and Clinical applications of Avaleha and Khanda kalpana

5 obj · 16 h · ceiling Does

2.1 SOP of Avaleha preparation with suitable Leha yogas
2.2 Therapeutic application of different varieties of Avaleha yogas
2.3 SOP of Khanda kalpana with suitable Khanda Yogas
2.4 Therapeutic application of different varieties of Khanda yogas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the therapeutic significance of Avaleha kalpana.	Lecture	1	Knows how
Discuss therapeutic significance of Paka and Khanda kalpana.	Lecture	1	Knows how
Discuss modern semisolid dosage forms.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the role of pharmaceutical additives and Samyak Paka of Avaleha.	Practical Training	5	Shows how
DOES — 1			
Analyse the role of selected leha yoga in different clinical conditions.	Experiential Learning	8	Does

Unit 3 · SOP of Sneha Kalpana and its clinical significance

5 obj · 17 h · ceiling Shows how

3.1 Different categories of Sneha with special reference to therapeutic application
3.2 Relevance of Avarthita Taila, Taila murchana and Taila Paka with special reference to critical endpoints
3.3 SOP of Sneha kalpana in selected formulations
3.4 Therapeutic application of selected Ghrita yogas and Taila yogas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the concept of Sneha Kalpana with special reference to critical end points (Sneha Paka) and its pharmaceutical and therapeutical significance.	Lecture	1	Knows how
Discuss and compare the significance of Murchana of Katu Taila, Tila Taila, Eranda Taila and Ghrita with special reference to analytical profile and pharmacotherapeutics.	Lecture	1	Knows how
Identify the similarities of Sneha Kalpa with liposomal drug delivery system.	Lecture	1	Knows how

SHOWS HOW — 2			
Demonstrate S.O.P for a selected Sneha Kalpa which is widely used and determine its analytical profile.	Practical Training	6	Shows how
Analyze different stages of Sneha Paka.	Experiential Learning	8	Shows how

Unit 4 · SOP and clinical applications of Arka and Dravakam in various disorders

5 obj · 14 h · ceiling Does

4.1 SOP for preparing selected Arkas
4.2 Analytical profiling of various yogas from Arka Kalpana
4.3 SOP of preparing various Dravakas
4.4 Analytical Profiling of selected formulations of Dravaka
4.5 Therapeutic applications and method of administration of Dravaka

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Desribe therapeutic significance of Arka Kalpana and Dravaka.	Lecture	1	Knows how
Discuss different distillation techniques used in pharmacy practice.	Lecture	1	Knows how
Elaborate Dravaka Kalpa with a critical correlation with acidic and alkaline preparations.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate S.O.P for the preparation of a selected Arka and Dravaka and create their analytical profile.	Practical Training	5	Shows how
DOES — 1			
Illusrate the clinical application of selected Arka/Dravaka Kalpa.	Experiential Learning	6	Does

Unit 5 · SOP and therapeutic applications of Sandhana Kalpana

6 obj · 17 h · ceiling Does

5.1 Different and novel fermentation techniques
5.2 SOP of preparing different types of Formulations in Sandhana Kalpana
5.3 Therapeutic applications of selected yogas from Sandhana Kalpana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe chemistry behind classical fermentation techniques.	Lecture	1	Knows how
Discuss industrial fermentation techniques.	Lecture	1	Knows how
Elaborate therapeutic significance of Sandhana kalpana.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate S.O.P for preparing a selected sandhana Yoga and determine the analytical standards.	Practical Training	5	Shows how
Record the dose, duration, indications, and clinical outcome of the various formulation i.e.Sandhana Kalpanas in the prescribed patients.	Experiential Learning	4	Shows how
DOES — 1			
Perform Comparative analysis of Asava Arishta Samples having different percentage and types of sugar.	Experiential Learning	5	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **11 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 29 · SEMESTER 5 · PAPER 4

Pharmacy Practice

2 credits | 60 notional hours | 3 units | **18 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	6	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Appraise about Basic Health Screening at community setup; Appraise and. Develop Standard Therapeutic Guidelines.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate Drug Information and Case Analysis; Demonstrate Prescription Handling and Dispensing; Identify Adverse drug reaction in selected case; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Describe Community Pharmacy; Discuss the Organization and structure of retail and wholesale drug stores; Describe the Drug distribution system in a hospital; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Define and classify Hospital Pharmacy, and discuss its organization.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Hospital Pharmacy and Community Pharmacy

7 obj · 26 h · ceiling Does

1.1 Hospital and its organization
1.2 Definition, Classification of Hospital
1.3 Hospital pharmacy and its organization
1.4 Functions of hospital pharmacy
1.5 Community Pharmacy
1.6 Organization and structure of retail and wholesale drug stores

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Define and classify Hospital Pharmacy, and discuss its organization.	Lecture	1	Knows
KNOWS HOW — 2			
Describe Community Pharmacy.	Lecture	1	Knows how
Discuss the Organization and structure of retail and wholesale drug stores.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate Drug Information and Case Analysis.	Practical Training	5	Shows how
Demonstrate Prescription Handling and Dispensing.	Practical Training	5	Shows how
Identify Adverse drug reaction in selected case.	Experiential Learning	7	Shows how
DOES — 1			
Appraise about Basic Health Screening at community setup.	Experiential Learning	6	Does

Unit 2 · Therapeutic Drug Monitoring

7 obj · 17 h · ceiling Shows how

2.1 Drug distribution system in a hospital
2.2 Dispensing of drugs to inpatients, types of drug distribution systems, charging policy and labelling, dispensing of drugs to ambulatory patients, and dispensing of
2.3 Hospital formulary
2.4 Therapeutic drug monitoring
2.5 Factors to be considered during the Therapeutic Drug Monitoring
2.6 Medication adherence
2.7 Patient medication history interview
2.8 Community pharmacy management
2.9 Financial, materials, staff, and infrastructure requirements

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe the Drug distribution system in a hospital.	Lecture	1	Knows how
Discuss Dispensing of drugs to inpatients, types of drug distribution systems, charging policy, and labelling, dispensing of drugs to ambulatory patients, and dispensing of controlled drugs.	Lecture	1	Knows how
Discuss about Hospital formulary and Therapeutic drug monitoring.	Lecture	1	Knows how
Evaluate Medication Adherence.	Lecture	1	Knows how
Analyse Financial, materials, staff, and infrastructure requirements.	Lecture	1	Knows how
SHOWS HOW — 2			
Perform Data Interpretation & Pharmacokinetic Calculations.	Practical Training	5	Shows how
Discuss Multidisciplinary Integration for Therapeutic Drug Monitoring.	Experiential Learning	7	Shows how

Unit 3 · Pharmacy and therapeutic committee

4 obj · 13 h · ceiling Does

3.1 Introduction to Pharmacy and Therapeutic Committee 3.2 Drug information services

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe Pharmacy and Therapeutic Committee along with Drug Information Services.	Lecture	1	Knows how
Enlist emergency drugs used in pharmacy.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate Formulary Review and Management.	Practical Training	5	Shows how
DOES — 1			
Appraise and. Develop Standard Therapeutic Guidelines.	Experiential Learning	6	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 30 · SEMESTER 5 · PAPER 4

Pharmaceutical Microbiology

2 credits | 60 notional hours | 3 units | **18 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	5	3
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Perform Simulation for Aseptic Practices; Conduct Real-life case studies on spoilage outbreaks and preservative failures; Observe and detect Spoilage of Pharmaceutical Product in Real Time.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
5 objectives

Demonstrate dry heat sterilization method; Conduct dry heat sterilization; Demonstrate the Aseptic Technique Practice for the transfer of microbes; and 2 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Discuss the beneficial and harmful roles of microbes in health, industry, and the environment; Describe the concept of sterilization, its importance, and differentiate various sterilization methods; Define contamination and explain its significance in aseptic and cleanroom environments; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Define and classify microorganisms (microbes) and explain their general characteristics.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to microbes with applications of sterilization

5 obj · 12 h · ceiling Shows how

1.1 Identification of bacteria using staining techniques (simple, Gram's & Acid-fast

1.2 Study of principles, procedures, merits, demerits, and applications of physical, chemical, gaseous, radiation, and mechanical methods of sterilization

1.3 Evaluation of the efficiency of sterilization methods

1.4 Equipment employed in large-scale sterilization and Sterility indicators

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Define and classify microorganisms (microbes) and explain their general characteristics.	Lecture	1	Knows
KNOWS HOW — 2			
Discuss the beneficial and harmful roles of microbes in health, industry, and the environment.	Lecture	1	Knows how
Describe the concept of sterilization, its importance, and differentiate various sterilization methods.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate dry heat sterilization method.	Practical Training	3	Shows how
Conduct dry heat sterilization.	Experiential Learning	6	Shows how

Unit 2 · Sources of contamination

6 obj · 12 h · ceiling Does

2.1 Designing an aseptic area

2.2 Study of different sources of contamination in an aseptic area and methods of prevention, clean area classification

2.3 Principles and methods of different microbiological assay methods

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Define contamination and explain its significance in aseptic and cleanroom environments.	Lecture	1	Knows how
Recognise common sources of contamination in pharmaceutical and microbiological settings and discuss each source contributes to microbial and particulate contamination.	Lecture	1	Knows how
Discuss the impact of contamination and describe methods used to minimize or prevent contamination.	Lecture	1	Knows how
Discuss the principles and methods of different microbiological assay methods.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the Aseptic Technique Practice for the transfer of microbes.	Practical Training	2	Shows how
DOES — 1			
Perform Simulation for Aseptic Practices.	Experiential Learning	6	Does

Unit 3 · Types of spoilage and use of antimicrobial agents

7 obj · 32 h · ceiling Does

3.1 Types of spoilage, factors affecting the microbial spoilage of pharmaceutical products,

3.2 Preservation of pharmaceutical products using antimicrobial agents, evaluation of microbial stability of formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Define spoilage, explain its significance in pharmaceuticals, food, and cosmetics, and discuss the factors that contribute to spoilage.	Lecture	1	Knows how
Describe the signs of spoilage.	Lecture	1	Knows how
Classify different types of antimicrobial agents (e.g., preservatives, disinfectants antibiotics) based on their mode of action and explain their antimicrobial mechanisms.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate Personnel Contamination Assessment.	Practical Training	7	Shows how
Demonstrate Different Types of Spoilage.	Practical Training	8	Shows how
DOES — 2			
Conduct Real-life case studies on spoilage outbreaks and preservative failures.	Experiential Learning	7	Does
Observe and detect Spoilage of Pharmaceutical Product in Real Time.	Experiential Learning	7	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 7 · SEMESTER 6 · PAPER 1

Sudha Varga, Sikata Varga, Visha Varga Dravyas

3 credits | 90 notional hours | 3 units | **22 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	10	10	2
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Perform Shodhana of Visha Dravyas as per different classics; Perform analytical testing process for Shodhita Visha dravya.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
10 objectives

Demonstrate Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya; Demonstrate the quality testing procedures for Bhasma/Pishti of Sudha vargeeya Dravya; Perform Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya; and 7 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
10 objectives

Discuss the list of Sudha varga Dravya mentioned in various classics and their mineralogical correlation; Discuss the pharmacological and therapeutic applications of Sudha Varga Dravya; Discuss research updates on characterization, pharmaceutical processing, QC, and pharmacotherapeutics of Sudhavarga Dravya; and 7 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Characterization, Pharmaceutical Processing, and Clinical Applications of Sudha Varga Dravya

7 obj · 28 h · ceiling Shows how

1.1 Classification and Identification of Sudha
1.2 Pharmaceutical Processing of Sudha Varga Dravya
1.3 Quality Control Methods for Sudha Varga Dravya
1.4 Pharmacology, Therapeutic Applications, and Research Updates on Sudha Varga Dravya

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the list of Sudha varga Dravya mentioned in various classics and their mineralogical correlation.	Lecture	1	Knows how
Discuss the pharmacological and therapeutic applications of Sudha Varga Dravya.	Lecture	2	Knows how
Discuss research updates on characterization, pharmaceutical processing, QC, and pharmacotherapeutics of Sudhavarga Dravya.	Lecture	2	Knows how
SHOWS HOW — 4			
Demonstrate Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya.	Practical Training	6	Shows how
Demonstrate the quality testing procedures for Bhasma/Pishti of Sudha vargeeya Dravya.	Practical Training	4	Shows how
Perform Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya.	Experiential Learning	7	Shows how
Perform quality testing procedures for Bhasma/Pishti of Sudha Vargeeya Dravya.	Experiential Learning	6	Shows how

Unit 2 · Characterization, pharmaceutical processing, and clinical applications of Sikata Varga Dravya

7 obj · 28 h · ceiling Shows how

2.1 Identification of Sikata Varga dravyas
2.2 Shodhana, marana, and other pharmaceutical processes of Sikata varga Dravya as per the SOP laid down
2.3 Quality control methods for the drugs of Sikata Varga Dravya
2.4 Pharmacology, therapeutic application, and research updates on Sikata Varga Dravya)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the list of Sikata varga Dravya mentioned in various classics and their mineralogical correlation.	Lecture	1	Knows how
Describe the pharmacology and therapeutic applications of Sikata Varga Dravya.	Lecture	2	Knows how
Discuss research updates on characterization, pharmaceutical processing, QC, and pharmacotherapeutics of Sikata Varga Dravya.	Lecture	2	Knows how
SHOWS HOW — 4			
Demonstrate Shodhana, marana, and Pishti Nirmana of Sikata varga Dravya.	Practical Training	6	Shows how
Demonstrate quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya.	Practical Training	4	Shows how
Perform Shodhana, marana and Pishti Nirmana of Sikata varga Dravya.	Experiential Learning	7	Shows how
Perform quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya.	Experiential Learning	6	Shows how

Unit 3 · Pharmaceutical processing, phytopharmacology and clinical applications of Visha Varga dravya

8 obj · 28 h · ceiling Does

3.1 Drugs of Visha and Upavisha categories
3.2 Shodhana of Visha-Upavisha Dravya based on different references
3.3 Phytochemical changes after the process of Shodhana
3.4 Clinical application and therapeutics of Visha Varga

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Discuss and compare the drugs of the Visha and Upavisha categories from different classics.	Lecture	1	Knows how
Describe the identification of herbal, animal, and mineral drugs mentioned under the Visha-Upavisha category based on organoleptic and physico-chemical characters.	Lecture	1	Knows how
Discuss the phytopharmacology of herbal drugs mentioned under the Visha-Upavisha category.	Lecture	2	Knows how
Describe formulations and therapeutic applications of Visha Dravya.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the Shodhana Process of Visha Dravya.	Practical Training	6	Shows how
Demonstrate the analytical testing process for Shodhita Visha dravya.	Practical Training	4	Shows how
DOES — 2			
Perform Shodhana of Visha Dravyas as per different classics.	Experiential Learning	7	Does
Perform analytical testing process for Shodhita Visha dravya.	Experiential Learning	6	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **12 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 8 · SEMESTER 6 · PAPER 1

Appraisal of Ratna and Uparatna and therapeutic significance of Pishti and Druti Kalpana

1 credits | 30 notional hours | 2 units | **13 objectives** | 25 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	7	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Identify substitutes for Ratnavarga and Uparatna varga drugs.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate mineralogical identification of precious and semiprecious stones under Ratna and Uparatna varga; Demonstrate SOP of Pishti Kalpana of Ratna-Uparatna; Identify the precious and semiprecious stones under Ratna and Uparatna varga using keys for mineralogical identification; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Discuss the characteristics of precious stones with special reference to the mineralogical identification of Ratnas; Describe different semiprecious stones mentioned in Rasashastra texts under the category of Uparatna; Discuss crystal chemistry, pharmacology and therapeutic applications of Ratna and Uparatnas; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Mineralogy, Crystal chemistry,and Pharmacotherapeutics of Ratna-Uparatna Varga

7 obj · 15 h · ceiling Does

1.1 Precious stones
1.2 Uparatna and semiprecious stones
1.3 Crystal chemistry and mineralogical properties of Ratna-Uparatna
1.4 Pharmacology and therapeutic applications of Ratna and Uparatnas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the characteristics of precious stones with special reference to the mineralogical identification of Ratnas.	Lecture	1	Knows how
Describe different semiprecious stones mentioned in Rasashastra texts under the category of Uparatna.	Lecture	1	Knows how
Discuss crystal chemistry, pharmacology and therapeutic applications of Ratna and Uparatnas.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate mineralogical identification of precious and semiprecious stones under Ratna and Uparatna varga.	Practical Training	3	Shows how
Demonstrate SOP of Pishti Kalpana of Ratna-Uparatna.	Practical Training	3	Shows how
Identify the precious and semiprecious stones under Ratna and Uparatna varga using keys for mineralogical identification.	Experiential Learning	3	Shows how
DOES — 1			
Identify substitutes for Ratnavarga and Uparatna varga drugs.	Experiential Learning	3	Does

Unit 2 · SOP and therapeutic applications of Pishti, Gandhaka Druti and Gandhaka Taila kalpana

6 obj · 13 h · ceiling Shows how

2.1 Pishti Kalpana: Concept, SOP, and Therapeutic
2.2 Classical concept of Druti
2.2 Gandhaka Druti and Gandhaka Taila Kalpana: Concept, SOP, and Therapeutic Applications
2.3 Pharmaceutical Modifications of Gandhaka Druti and Gandhaka Taila

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the concept, significance, and therapeutic value of Pishti kalpas.	Lecture	1	Knows how
Discuss the concept of Druti mentioned in Rasa classics and explore the therapeutic value of Gandhaka Druti and Gandhaka Taila.	Lecture	1	Knows how
SHOWS HOW — 4			
Demonstrate different methods of Gandhaka Druti.	Practical Training	2	Shows how
Demonstrate Pharmaceutical modification of Gandhaka Druti and Gandhaka Taila.	Practical Training	2	Shows how
Prepare Pishti of Selected Ratna or Uparatna.	Experiential Learning	3	Shows how
Prepare Gandhaka Druti using different media.	Experiential Learning	4	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 15 · SEMESTER 6 · PAPER 2

Disease condition specific selection of formulation

2 credits | 60 notional hours | 5 units | **12 objectives** | 50 formative marks | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	6	6	—
not used	certified here	certified here	not used

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	6	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	3	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
not used here

This module does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate the essential formulations needed for the diverse branches of Ayurveda; Identify different dosage forms used by local health practitioners. through filed based survey; Analyze the key characteristics of dosage forms that meet the needs of various routes of drug administration; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
6 objectives

Appraise the rational relationship between Roga and Bhaishajya Kalpana based on classical references; Analyse and justify the selection of formulation according to the Rogavastha (stage of disease); Describe different branches / specialty practices in Ayurveda and justify the need for specific dosage forms for effective clinical outcomes; and 3 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this module cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Application of Yukti Pramana in Formulation Selection

2 obj · 2 h · ceiling Knows how

1.1 Disease Category and Determination of multiple factors involved in diagnosis and
1.2 Strategic selection of drugs and dosage forms
1.3 Stage-specific selection (Rogavastha and Bhaishajya Kalpana)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Appraise the rational relationship between Roga and Bhaishajya Kalpana based on classical references.	Lecture	1	Knows how
Analyse and justify the selection of formulation according to the Rogavastha (stage of disease).	Lecture	1	Knows how

Unit 2 · Selection of formulations for Speciality practice in Ayurveda

3 obj · 12 h · ceiling Shows how

2.1 Speciality practice in Ayurveda
2.2 Alignment of formulations with the specialty branch of clinical Practice
2.3 Drugs with ideal guna, virya, vipaka, and of appropriate dosage forms for the patient and pathology
2.4 Evidence-based drugs and dosage form selection for specialty practice

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe different branches /specialty practices in Ayurveda and justify the need for specific dosage forms for effective clinical outcomes.	Lecture	2	Knows how
SHOWS HOW — 2			
Demonstrate the essential formulations needed for the diverse branches of Ayurveda.	Practical Training	5	Shows how
Identify different dosage forms used by local health practitioners. through filed based survey.	Practical Training	5	Shows how

Unit 3 · Routes of Drug Administration

2 obj · 12 h · ceiling Shows how

3.1 Classical and conventional concepts of Drug Administration
3.2 Routes of Drug Administration – Need and Relevance
3.3 Factors in the selection of drug and dosage form based on route of administration
3.4 Matching Dosage Forms to Pharmacological Needs by Selected Route
3.5 Properties of formulations and their effect on Pharmacodynamics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Analyze the formulations and routes of drug administration, and justify the relevance of each route in pharmacology.	Lecture	2	Knows how
SHOWS HOW — 1			
Analyze the key characteristics of dosage forms that meet the needs of various routes of drug administration.	Experiential Learning	10	Shows how

Unit 4 · Designing formulations for Shodhana and Shamana Chikitsa

2 obj · 12 h · ceiling Shows how

4.1 Ideal characteristics of drugs as per classics (Matravat Bheshaja lakshanani)
4.2 Principles of Shodhana chikitsa (Purificatory/detoxifying therapy) and actions of formulations used for shodhana - Langhana, Doshanirhana, srotahshodhana
4.3 Principles of Shamana chikitsa (palliative treatment) and actions of formulations used for Shamana- Deepana, Pachana, Stambhana, Rasayana, etc
4.4 Evidence-based dosage forms and formulations for common diseases

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Compare dosage forms and formulations used for shodhana and shamana chikitsa.	Lecture	2	Knows how

SHOWS HOW — 1			
Demonstrate the method of prepartion of frequently used formulation for Shodhana and Shamana Chikitsa of common diseases.	Practical Training	10	Shows how

Unit 5 · Innovation in Ayurvedic Formulations

3 obj · 18 h · ceiling Shows how

5.1 Need and current trends in the industry of Ayurvedic Pharmaceutics

5.2 Innovative Formulations-Transdermal patches, Nebulizers, Eye masks, etc

5.3 Relevance and consequences of innovation done by the R&D department of Ayurvedic Pharmacies

5.4 Knowledge, Attitude, and Practice (KAP) studies on various aspects related to Ayurvedic Formulations available in the market

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss challenges associated with recent modifications and innovations in Ayurvedic formulations, specifically focusing on transdermal patches and nebulizers.	Lecture	2	Knows how
SHOWS HOW — 2			
Demonstrate the method of nnovative formulations that enhance well-being and redefine healthcare.	Experiential Learning	10	Shows how
Demonstrate the essential competencies needed to develop R&D department of Ayurvedic Pharmacy.	Experiential Learning	6	Shows how

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **6 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Shows how.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 16 · SEMESTER 6 · PAPER 2

Current trends in Ayurvedic Pharmaceutical Industry.

2 credits | 60 notional hours | 3 units | **12 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	6	4	1
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	6	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	3	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Assess real-life experience of the market for the growing popularity of Ayurvedic products.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
4 objectives

Analyze the Market Status of Ayurvedic Pharmacy Industry; Interpret benefits of field visit and Internship; Demonstrate and assist quality control practical to display quality standards of Ayurvedic formulations; and 1 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
6 objectives

Discuss key competencies required for employment in the Ayurvedic pharmacy; Describe career opportunities in Ayurvedic pharma industries; Discuss key competencies required for employment in the Ayurvedic pharmacy; and 3 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Discuss current trends in Ayurvedic pharma industry.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Key Competencies Required for Employment in Pharmacy

6 obj · 22 h · ceiling Does

1.1 Orientation about current trends in the Ayurvedic Pharmaceutical Industry
1.2 Ayurvedic Pharmacy Industry (India & Global Overview): Market size, growth, and key importing countries
1.3 Ayurvedic Products and Market Stakeholders
1.4 Consumer Preferences and Distribution Channels

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss current trends in Ayurvedic pharma industry.	Lecture	1	Knows
KNOWS HOW — 3			
Discuss key competencies required for employment in the Ayurvedic pharmacy.	Lecture	1	Knows how
Describe career opportunities in Ayurvedic pharma industries.	Lecture	1	Knows how
Discuss key competencies required for employment in the Ayurvedic pharmacy.	Lecture	1	Knows how
SHOWS HOW — 1			
Analyze the Market Status of Ayurvedic Pharmacy Industry.	Practical Training	8	Shows how
DOES — 1			
Assess real-life experience of the market for the growing popularity of Ayurvedic products.	Experiential Learning	10	Does

Unit 2 · Internship/ Visits to GMP certified Pharmacies

3 obj · 16 h · ceiling Shows how

2.1 Factory set up and daily Work flow
2.2 Hygiene checks & Cleaning
2.3 Inventory Management, material management and Production
2.4 Packaging and labelling
2.5 Documentation (batch records) and QC record

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the sections and working flow of GMP-certified pharmacy.	Lecture	2	Knows how
Observe and appraise the operational management of Ayurvedic Pharmacy.	Practical Training	6	Knows how
SHOWS HOW — 1			
Interpret benefits of field visit and Internship.	Experiential Learning	8	Shows how

Unit 3 · Quality Assurance and Quality Control (QAQC) Laboratory Training

3 obj · 18 h · ceiling Shows how

3.1 Information about Leading Ayurvedic Research Labs in India
3.2 Regulatory & Certification Bodies for QAQC of Ayurvedic Formulations
3.3 Laboratory set up and daily Work flow
3.4 LIMS (Laboratory Information Management System)
3.5 Instrument Calibration, Sample testing and analysis, Record keeping

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe modern parameters used for quality assurance and quality control of Ayurvedic Drugs.	Lecture	4	Knows how
SHOWS HOW — 2			
Demonstrate and assist quality control practical to display quality standards of Ayurvedic formulations.	Practical Training	6	Shows how
Justify the utility of LIMS(Laboratory Information Management System) in Ayurvedic research laboratories and the workflow of the laboratory.	Experiential Learning	8	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **5 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 23 · SEMESTER 6 · PAPER 3

SOP and clinical applications of Churna Kalpana(Powder dosage forms)

3 credits | 90 notional hours | 5 units | **26 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	14	7	4
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	15	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
4 objectives

Evaluate the role of a selected Churna in a specific clinical condition; Demonstrate the preparation of a selected Paneeya Kshara and Pratisaraneeya Kshara and their quality analysis; Demonstrate the preparation of a selected Lavana formulation following different methods and analytical profiling; and 1 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate the SOP and quality analysis of a selected Churna yoga; Demonstrate the therapeutic applications of Kshara and Ksharasutra in a given selected; Demonstrate the role of Lavana Kalpana in various clinical conditions; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
14 objectives

Discuss the importance of particle size of different types of Churna concerning its pharmaceutical and therapeutic significance; Describe the pharmacological and therapeutic significance of Churna kalpana; Discuss Kshara Kalpana, its sources, and evaluate the chemical composition; and 11 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Discuss SOP for the preparation of Masi by following different classical and advanced methods IS.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · SOP and clinical utility of Churna Kalpana

4 obj · 15 h · ceiling Does

1.1 Standardisation of various Churna yogas with Standard Operating Procedure
1.2 Therapeutic application of various Churna yogas
1.3 Relevance of Churna kalpana in therapeutics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the importance of particle size of different types of Churna concerning its pharmaceutical and therapeutic significance.	Lecture	1	Knows how
Describe the pharmacological and therapeutic significance of Churna kalpana.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the SOP and quality analysis of a selected Churna yoga.	Practical Training	5	Shows how
DOES — 1			
Evaluate the role of a selected Churna in a specific clinical condition.	Experiential Learning	8	Does

Unit 2 · Standardization, analytical profiling, and clinical application of Kshara

5 obj · 16 h · ceiling Does

2.1 Relevance of Kshara kalpana and Kshara sutra in therapeutics
2.2 SOP of preparation of Paneeya Kshara and Pratisaraneeya Kshara
2.3 SOP of preparation of Kshara Sutra
2.4 Analytical profiling of Kshara

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss Kshara Kalpana, its sources, and evaluate the chemical composition.	Lecture	1	Knows how
Describe the therapeutic application of Paneeya Kshara and Prathisaraneeya Kshara.	Lecture	1	Knows how
Discuss the techniques for the preparation of Kshara Sutra.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the therapeutic applications of Kshara and Ksharasutra in a given selected.	Experiential Learning	8	Shows how
DOES — 1			
Demonstrate the preparation of a selected Paneeya Kshara and Pratisaraneeya Kshara and their quality analysis.	Practical Training	5	Does

Unit 3 · Lavana Kalpana with special reference to pharmaceutico-analytical aspects and therapeutic attributes

5 obj · 16 h · ceiling Does

3.1 Standard Operating
3.2 Analytical Profiling of Lavanas
3.3 Therapeutic Applications of Selected Formulations in Lavana Kalpana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe salt-based formulations (Lavana Kalpa) mentioned in different texts.	Lecture	1	Knows how
Discuss chemical profiling and standardization of Lavana Kalpa.	Lecture	1	Knows how
Illustrate the probable mechanism of action and clinical significance of Lavana Kalpa.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the role of Lavana Kalpana in various clinical conditions.	Experiential Learning	8	Shows how

DOES — 1			
Demonstrate the preparation of a selected Lavana formulation following different methods and analytical profiling.	Practical Training	5	Does

Unit 4 · Satva Kalpana SOP and clinical significance

5 obj · 15 h · ceiling Shows how

4.1 SOP for preparing selected Satvas
4.2 Analytical profiling of various yogas from Satva Kalpana
4.3 Therapeutic application of Satva kalpana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the uniqueness and pharmaceutical relevance of Satva kalpana.	Lecture	1	Knows how
Discuss SOP and analytical profiling of Satva Kalpana.	Lecture	1	Knows how
Evaluate the clinical significance of Satva Kalpana.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the preparation of a selected Satva and perform its analytical profiling.	Practical Training	5	Shows how
Demonstrate the role of a selected satva in a specified disease condition.	Experiential Learning	7	Shows how

Unit 5 · Therapeutic attributes of Masi Kalpana

7 obj · 22 h · ceiling Does

5.1 Masi from plant and animal origin
5.2 Pharmacology of Masi
5.3 Activated charcoal in therapeutics
5.4 Clinical applications of Masi

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss SOP for the preparation of Masi by following different classical and advanced methods IS.	Lecture	1	Knows
KNOWS HOW — 3			
Discuss the analytical profiling of Mashi Kalpas.	Lecture	1	Knows how
Discuss the therapeutic significance of various Masi Kalpas.	Lecture	1	Knows how
Evaluate the clinical significance of activated charcoal.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the preparation of Mashi and interpret its analytical profiling.	Practical Training	5	Shows how
Demonstrate pharmaceutical modification of Masi Kalpana.	Practical Training	5	Shows how
DOES — 1			
Evaluate the Role of Malaharas as a Topical application in Skin disease.	Experiential Learning	8	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **11 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 24 · SEMESTER 6 · PAPER 3

Therapeutic significance of Topical dosage forms

1 credits | 30 notional hours | 3 units | **11 objectives** | 25 formative marks | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	6	—
not used	certified here	certified here	not used

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	3	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
not used here

This module does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate various Netra Kalpa and their clinical applications; Demonstrate the role of Ayurvedic ophthalmic preparations in various eye diseases; Prepare a novel dosage form using one of the classical lepa Yoga; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Discuss classical and novel dosage forms for ophthalmic, otic, and buccal use; Describe therapeutic significance and clinical applications of dosage forms for ophthalmic, otic, and buccal use; Discuss classical topical dosage forms like Lepa, Malahara; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this module cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Ophthalmic, Otic, and Buccal dosage forms and their clinical applications1.SOP for Ophthalmic, Otic and buccal dosage forms

4 obj · 10 h · ceiling Shows how

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss classical and novel dosage forms for ophthalmic, otic, and buccal use.	Lecture	1	Knows how
Describe therapeutic significance and clinical applications of dosage forms for ophthalmic, otic, and buccal use.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate various Netra Kalpa and their clinical applications.	Practical Training	3	Shows how
Demonstrate the role of Ayurvedic ophthalmic preparations in various eye diseases.	Experiential Learning	5	Shows how

Unit 2 · Standardisation of Lepa and Malahara preparations and its application in various disorders

3 obj · 6 h · ceiling Shows how

2.1 SOP for the preparation of various Lepas
2.2 Innovative approaches including Patches, Bandages, Bands
2.3 Clinical applications of Lepa and Malahara kalpa

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss classical topical dosage forms like Lepa, Malahara.	Lecture	1	Knows how
Describe novel dosage forms for topical application, along with their regulatory guidelines.	Lecture	1	Knows how
SHOWS HOW — 1			
Prepare a novel dosage form using one of the classical lepa Yoga.	Practical Training	4	Shows how

Unit 3 · Standard Operating Procedure (SOP) for Upanaha, Dhupa, Dhumapana, and Avachurnana Kalpa and their therapeutic applications

4 obj · 12 h · ceiling Shows how

3.2 Anti-microbial activity of Dhupa
3.5 Clinical application of various Dhumapana yogas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss the therapeutic significance of Upanaha, Dhupa, Dhumapana.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the preparation of a selected Upanaha which is useful in treating Shopha.	Practical Training	3	Shows how
Demonstrate the role of lepa Yogas in managing various medical needs by innovative approaches.	Experiential Learning	5	Shows how
Demonstrate the role of Malahara as a Topical application in Skin disease.	Experiential Learning	3	Shows how

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **6 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Shows how.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 31 · SEMESTER 6 · PAPER 4

Regulatory Acts and Rules applicable to Ayurvedic Drugs

3 credits | 90 notional hours | 6 units | **31 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	13	8	8
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	15	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	7	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	9	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
8 objectives

Analyse case law on the Factory Act; Prepare a technical report on Adverse Drug Reactions (ADRs) for submission to DTAB; Identify misbranded and spurious drugs/cosmetics; and 5 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
8 objectives

Record their observations regarding hypothetical issue through Mock composition exercise of demonstration of the roles of DTAB (Drug Technical Advisory Board) composition, including representatives from DGHS, DCGI, pharmacologists; Prepare mock import license applications; Demonstrate case scenarios of misbranded/spurious drugs and cosmetics; and 5 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
13 objectives

Describe the Scope and Applicability of the Drugs and Cosmetics Act and Key Definitions; Discuss the Legal importance of Drug Regulation and enlist prescribed books of ASU from the first Schedule of D&C Act; Identify key regulatory authorities and develop foundational knowledge of regulatory frameworks; and 10 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Summarize the provisions of Chapter III regarding the import of drugs and cosmetics; Discuss Medicinal and Toilet Preparations Act, 1955, Opium Act, 1857, and Pharmacy.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Chapter I of the Drugs and Cosmetics Act, 1940 and Rules 1945

3 obj · 6 h · ceiling Does

1.1 Introduction to 'Drugs and Cosmetic Act, 1940; and 1945' and definitions

1.2 Prescribed Books on Ayurveda, Siddha, and Unani Drugs: The First Schedule

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the Scope and Applicability of the Drugs and Cosmetics Act and Key Definitions.	Lecture	1	Knows how
Discuss the Legal importance of Drug Regulation and enlist prescribed books of ASU from the first Schedule of D&C Act.	Lecture	1	Knows how
DOES — 1			
Analyse case law on the Factory Act.	Experiential Learning	4	Does

Unit 2 · Chapter II - Regulatory Bodies

3 obj · 9 h · ceiling Does

2.1 The Drugs Technical Advisory Board (DTAB)

2.2 The Central Drugs Laboratory (CDL)

2.3 The Drugs Consultative Committee (DCC)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Identify key regulatory authorities and develop foundational knowledge of regulatory frameworks.	Lecture	1	Knows how
SHOWS HOW — 1			
Record their observations regarding hypothetical issue through Mock composition exercise of demonstration of the roles of DTAB (Drug Technical Advisory Board) composition, including representatives from DGHS, DCGI, pharmacologists.	Practical Training	4	Shows how
DOES — 1			
Prepare a technical report on Adverse Drug Reactions (ADRs) for submission to DTAB.	Experiential Learning	4	Does

Unit 3 · Chapters III, IV, and IV-A - Regulation of Drugs and Cosmetics

8 obj · 18 h · ceiling Does

3.1 Chapter III: Standards of quality

3.2 Misbranded, adulterated, and spurious drugs and cosmetics

3.3 Prohibition of import of certain drugs or cosmetics

3.4 Manufacturing, Labelling, and Sale of Drugs and Cosmetics (Chapter IV-A)

3.5 Standards of quality for drugs and cosmetics (Sections 16 and 16A)

3.6 Prohibition of manufacture and sale of certain drugs and cosmetics (Sections 18 and 19)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Summarize the provisions of Chapter III regarding the import of drugs and cosmetics.	Lecture	1	Knows
KNOWS HOW — 4			
Identify legal restrictions and prohibitions for misbranded and spurious drugs and.	Lecture	1	Knows how
Describe manufacturing, labelling, and sale provisions, including penalties and seizures (Chapter IV-A).	Lecture	1	Knows how
Justify the importance of standards of quality for drugs and cosmetics (Sections 16 and.	Lecture	1	Knows how
Discuss the prohibition of the manufacture and sale of certain drugs and cosmetics.	Lecture	1	Knows how

SHOWS HOW — 2			
Prepare mock import license applications.	Practical Training	4	Shows how
Demonstrate case scenarios of misbranded/spurious drugs and cosmetics.	Practical Training	5	Shows how
DOES — 1			
Identify misbranded and spurious drugs/cosmetics.	Experiential Learning	4	Does

Unit 4 · Schedules to Drugs and Cosmetics Rules related to ASU drugs and cosmetics

5 obj · 16 h · ceiling Does

4.1 Good Manufacturing Practices (GMP) for ASU Medicines

4.2 Prescription and Safety Requirements (Schedules H, G, X) and Pharmacological Information (Schedule Y)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the provision of Schedule T and Schedule E (1).	Lecture	1	Knows how
Discuss the requirements of Schedules H, G, X (prescription and safety), and Schedule Y (pharmacological information).	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the formulation of a proprietary formula and apply for a license.	Practical Training	5	Shows how
DOES — 2			
Apply for GMP certification.	Experiential Learning	4	Does
Prepare a new formula and apply for a mock license.	Experiential Learning	5	Does

Unit 5 · Pharmaceutical observation of drugs and cosmetics ASU drugs

3 obj · 10 h · ceiling Does

5.1 Shelf life and stability of ASU drugs: recent amendments

5.2 Use of Preservatives in ASU drugs

5.3 Packaging and labelling requirements for ASU drugs

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe Shelf life of ASU drugs: recent amendments, Preservatives, and. Discuss Packaging and labelling technique.	Lecture	1	Knows how
DOES — 2			
Evaluate label compliance and identify shortcomings in given labels.	Experiential Learning	5	Does
Design and develop a label for a hypothetical ASU product, ensuring compliance with regulatory requirements.	Experiential Learning	4	Does

Unit 6 · Miscellaneous Regulatory Acts related to drugs and cosmetics

9 obj · 25 h · ceiling Does

6.1 Narcotic and Psychotropic Substances Act 1985

6.2 Factories Act, 1948

6.3 Industries Act, 1951

6.4 Biological Diversity Act 2002

6.5 Consumer Protection Act 1986

6.6 Drugs and Magic Remedies Act, 1954

6.7 Medicinal and Toilet Preparations Act, 1955

6.8 Opium Act, 1857

6.9 Pharmacy Act 1948

6.10 Recent Amendments in D & C Act concerning Ayurvedic Drug Industry

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss Medicinal and Toilet Preparations Act, 1955, Opium Act, 1857, and Pharmacy.	Lecture	1	Knows

KNOWS HOW — 3			
Describe the key provisions of the Narcotic and Psychotropic Substances Act, 1985 Factories Act, 1948, and Industries Act, 1951.	Lecture	1	Knows how
Discuss the Biological Diversity Act 2002, Consumer Protection Act 1986, and Drugs and Magic Remedies Act, 1954.	Lecture	1	Knows how
Describe Recent Amendments in D & C Act concerning Ayurvedic Drug Industry.	Lecture	1	Knows how
SHOWS HOW — 4			
Demonstrate Landmark Judgments under the NDPS Act.	Practical Training	4	Shows how
Demonstrate Case Law on the Factory Act.	Practical Training	4	Shows how
Demonstrate Case Study under the Drugs and Magic Remedies Act.	Practical Training	4	Shows how
Analyse Landmark Judgments under the NDPS Act.	Experiential Learning	5	Shows how
DOES — 1			
Prepare Case Studies and Report related to Objectionable Advertisements.	Experiential Learning	4	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **16 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 32 · SEMESTER 6 · PAPER 4

Entrepreneurship and Start-Ups

1 credits | 30 notional hours | 4 units | **14 objectives** | 25 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	4	3	5
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
5 objectives

Create an elevator pitch for a startup idea; Conduct and document market research for a start-up, including survey design and analysis; Carry out A Case Study-Based Role-Play, navigating a Startup's Early Days; and 2 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
3 objectives

Demonstrate the process of patent filing; Demonstrate a case scenario of the new molecule patent; Demonstrate a case scenario of patenting the formulation process.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Discuss the dynamism of Entrepreneurial Motivation and key entrepreneurial competencies; Demonstrate the process of development of a basic business model as per the template; Identify the challenges and feasibility report of the start-up; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Describe entrepreneurship, the startup life cycle, the MSME Act, and the process of registration; Identify the role of DPIIT (Department for Promotion of Industry and Internal Trade) BHASKAR (Bharat Startup Knowledge Access Registry) in the promotion of entrepreneurship and startups.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Entrepreneurship and Startup

3 obj · 4 h · ceiling Does

1.1 Introduction to Entrepreneurship and Start-ups

1.2 MSMED Act and its Process of Registration and Role of Department for Promotion of Industry and Internal Trade(DPIIT) and Bharat Startup Knowledge Access

1.3 Types of Enterprises and Role of Enterprises in National and Global Economy

1.4 Government Policies, Schemes, and Institutional Support for Entrepreneurship

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 2			
Describe entrepreneurship, the startup life cycle, the MSME Act, and the process of registration.	Lecture	1	Knows
Identify the role of DPIIT (Department for Promotion of Industry and Internal Trade) BHASKAR (Bharat Startup Knowledge Access Registry) in the promotion of entrepreneurship and startups.	Lecture	1	Knows
DOES — 1			
Create an elevator pitch for a startup idea.	Experiential Learning	2	Does

Unit 2 · Process of entrepreneurship development

3 obj · 8 h · ceiling Does

2.1 Need and the process of entrepreneurship development

2.2 Dynamics of Entrepreneurial Motivation

2.3 Developing Entrepreneurial Competencies

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the dynamism of Entrepreneurial Motivation and key entrepreneurial competencies.	Lecture	1	Knows how
Demonstrate the process of development of a basic business model as per the template.	Practical Training	3	Knows how
DOES — 1			
Conduct and document market research for a start-up, including survey design and analysis.	Experiential Learning	4	Does

Unit 3 · Project Proposal for new Invention/startup

2 obj · 3 h · ceiling Does

3.1 Challenges and the feasibility report of the Startups

3.2 Planning, resource mobilisation, and implementation of a start-up project

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Identify the challenges and feasibility report of the start-up.	Lecture	1	Knows how
DOES — 1			
Carry out A Case Study-Based Role-Play, navigating a Startup's Early Days.	Experiential Learning	2	Does

Unit 4 · Intellectual Property Rights and Patents

6 obj · 13 h · ceiling Does

4.1 Introduction to IPR and Patents

4.2 Indian and international patent laws, proposed amendments as applicable to herbal/natural products and processes

4.3 Procedure for patent filing

4.4 Geographical Indication and Copyright

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Define IPR, describe the purpose and importance of IPR, and the laws governing patent.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the process of patent filing.	Practical Training	3	Shows how
Demonstrate a case scenario of the new molecule patent.	Practical Training	2	Shows how
Demonstrate a case scenario of patenting the formulation process.	Practical Training	2	Shows how

DOES — 2			
Design an application to Patent the formulation process.	Experiential Learning	3	Does
Identify the patentability of a product.	Experiential Learning	2	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PART C

Reference

Four appendices, generated from the same data as Part B so the two cannot disagree. C1 decodes the tiers and names the instruments that certify each. C2 gives the ceiling of every module in one view — the fastest way to check that an assessment matches what it is meant to certify. C3 lists the instruments the curriculum itself names, sorted by the tier each can honestly reach. C4 records the sources and what was and was not altered.

1. **C1** The four tiers, and what certifies each
2. **C2** Ceiling of every module
3. **C3** The instruments this curriculum names, by tier
4. **C4** Sources, and a note on how this book was made

↑ Contents

PART C · APPENDIX C1

The four tiers, and what certifies each

Miller's four tiers with the count of objectives at each, the instruments that can certify them, and the evidence you keep. Nothing below a tier can certify it.

TIER	OBJECTIVES	INSTRUMENTS	EVIDENCE
Knows	21	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	267	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	222	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	67	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

TIER	OBJECTIVES	SHARE
Knows	21	3.6%
Knows how	267	46.3%
Shows how	222	38.5%
Does	67	11.6%

Total 577 objectives.

PART C · APPENDIX C2

Ceiling of every module

The highest tier each module reaches. Read across: the ceiling names the evidence you must gather, and the tier above it names what you may not ask for.

M	U	UNIT	OBJ	CEILING
SEMESTER 3				
MODULE 1 — CHRONOLOGICAL DEVELOPMENT AND TAXONOMY OF RASA DRAVYAS — 19 OBJECTIVES, CEILING DOES				
1	1	Chronological Development and Overview of Rasashastra	5	Shows how
1	2	Taxonomy of Rasadravyas	5	Shows how
1	3	Geological/mineralogical identification of Rasadravyas	6	Does
1	4	Overview of Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya	3	Shows how
MODULE 2 — PUTA AND SAMSKARA (TRANSFORMATIVE PROCESSES OF DRUG MANUFACTURING) — 21 OBJECTIVES, CEILING DOES				
2	1	Puta and Law of thermodynamics	6	Does
2	2	Concept, understanding, application, SOPs and chemical changes of Jarana, Amritikarana and Lohitikarana	4	Shows how
2	3	Concept, understanding, application, SOPs (Standard Operating Procedures) and chemical changes of Satva and Druti procedures	4	Shows how
2	4	Samskara (Principles of Transformative processes of drug manufacturing)	2	Shows how
2	5	Influence of Desha (place) and Kala(time and seasonal factors) of drug qualities	2	Shows how
2	6	Kalaprakarsha- Processing with/Effect of specific time exposure and Bhajana (processing vessel)	3	Shows how
MODULE 9 — SCIENTIFIC FUNDAMENTALS OF AYURVEDIC PHARMACEUTICAL PREPARATIONS — 15 OBJECTIVES, CEILING DOES				
9	1	Classical Principles of drug processing and scientific basis of Panchavidha Kashaya Kalpana	3	Does
9	2	Identity, Purity and Quality of Raw materials used in Ayurveda Pharmaceutics	3	Shows how
9	3	Pharmaceutical processing of drugs with its rationality	3	Shows how
9	4	Ayurvedic Classical Dosage Forms and their integration with modern scientific validation	6	Does
MODULE 10 — PRINCIPLES OF PHARMACEUTICAL FORMULATION — 12 OBJECTIVES, CEILING DOES				
10	1	Principles of Samskara (transformative processes of drug manufacturing) in relation to drug development	2	Shows how
10	2	Rationality behind Composition of Formulations	2	Shows how
10	3	Principles and designing of various dosage forms	2	Shows how
10	4	Principles of Prakshep dravyas (Pharmaceutical Additives)	3	Shows how
10	5	Principles of Potency and Efficacy enhancement	3	Does
MODULE 17 — GUIDELINES FOR ADMINISTRATION OF RASOUSHADHI IN CLINICAL PRACTICE — 22 OBJECTIVES, CEILING DOES				
17	1	Concept of Rasa chikitsa and significance of Rasoushadhi	4	Shows how
17	2	Concept of Kshetrikarana and its significance prior to administration of Rasoushadhi	4	Shows how
17	3	Pathya -Apathya during Rasayoga Sevana	5	Does

M	U	UNIT	OBJ	CEILING
17	4	Safety and toxicity of Rasoushadhi	5	Does
17	5	Heavy metal toxicity and chelation therapy	4	Does
MODULE 18 — PRAYOGIKA BHAISHAJYA KALPANA(CLINICAL APPLICATION OF PHARMACEUTICAL PRODUCTS) — 24 OBJECTIVES, CEILING DOES				
18	1	Aushadha Sevana Kala (Time of administration of drug)	7	Does
18	2	Concept and scientific basis of Anupana	6	Does
18	3	Matra (Posology)	5	Shows how
18	4	Concept of drug repurposing	6	Shows how
MODULE 25 — ARTIFICIAL INTELLIGENCE IN DRUG DEVELOPMENT AND PHARMACOGENETICS; PHARMACOGENOMICS — 20 OBJECTIVES, CEILING DOES				
25	1	Role of Artificial intelligence in drug development	4	Does
25	2	Types of software used in Artificial intelligence	3	Does
25	3	3D printed drug delivery system	5	Shows how
25	4	Pharmacogenetics and Pharmacogenomics	4	Shows how
25	5	Pharmacogenomics in drug discovery	4	Shows how
MODULE 26 — PHARMACOLOGY — 16 OBJECTIVES, CEILING DOES				
26	1	Introduction to Pharmacology	3	Knows how
26	2	Pharmacokinetics, therapeutic index, and drug interactions	3	Knows how
26	3	Pharmacology of drugs that act on the peripheral nervous system	3	Knows how
26	4	Pharmacology of drugs that act on the central nervous system I	3	Shows how
26	5	Pharmacology of drugs that act on the central nervous system II	4	Does
SEMESTER 4				
MODULE 3 — PARADA (MERCURY) AND ITS PROCESSING: ANCIENT WISDOM AND SCIENTIFIC ADVANCES — 17 OBJECTIVES, CEILING DOES				
3	1	Parada: Ancient and Contemporary Perspectives on Its Shodhana and Underlying Principles	8	Does
3	2	Parada Ashta Samskara and Ashtadasha Samskara	4	Shows how
3	3	Physico-chemical proprties and pharmacology of mercury and mercurial compounds	2	Shows how
3	4	Mercury toxicity, ADR and its management	3	Shows how
MODULE 4 — MAHARASA DRAVYA, ITS PROCESSING AND UTILITY — 22 OBJECTIVES, CEILING DOES				
4	1	Maharasa Dravyas with special reference to Abhraka and Vaikranta	8	Shows how
4	2	Principles and Processing of Makshika and Vimala	3	Shows how
4	3	Principles and Processing of Shilajit and Sasyaka	3	Shows how
4	4	Principles and Processing of Chapala and Kharpara, along with physico-chemical analysis, with research updates of all Maharasa Dravyas	4	Does
4	5	Utility of new metals and minerals	4	Shows how
MODULE 11 — MANA PARIBHASHA (METROLOGY IN AYURVEDIC PHARMACEUTICAL PREPARATION) — 16 OBJECTIVES, CEILING DOES				
11	1	Mana Paribhasha (Classical system of measurement)	4	Does
11	2	Standard Units of Measurement	3	Shows how

M	U	UNIT	OBJ	CEILING
11	3	Volumetric Vs Gravimetric Methods	3	Shows how
11	4	Rationality of Proportion of Ingredients in Ayurvedic formulations	3	Does
11	5	Bulk production of Ayurvedic Formulations	3	Does
MODULE 12 — PROCESS VALIDATION PROTOCOL — 10 OBJECTIVES, CEILING SHOWS HOW				
12	1	Pharmaceutical Preparation process: Classical Dosage forms	6	Shows how
12	2	Pharmaceutical Preparation process: Recent advances of Dosage forms	3	Shows how
12	3	Novel Dosage and drug delivery Forms	1	Shows how
MODULE 19 — SOP AND CLINICAL APPLICATION OF CHATURVIDHA RASAYANA — 25 OBJECTIVES, CEILING DOES				
19	1	Standard Operating Procedure (SOP) and Clinical Application of Khalveeya Rasayana	5	Shows how
19	2	SOP and clinical application of Parpati Rasayana	6	Does
19	3	SOP and clinical application of Pottali Rasayana	7	Does
19	4	SOP and clinical application of Kupipakwa Rasayana	7	Shows how
MODULE 20 — SOP AND CLINICAL APPLICATION OF AYASKRITI, LOUHA, AND MANDURA KALPA — 12 OBJECTIVES, CEILING DOES				
20	1	SOP and clinical application of Ayaskriti	3	Shows how
20	2	SOP and clinical application of Louha kalpa	4	Shows how
20	3	SOP and Clinical application of Mandura Kalpa	5	Does
MODULE 27 — NETWORK PHARMACOLOGY & REVERSE PHARMACOLOGY — 16 OBJECTIVES, CEILING DOES				
27	1	Netwok Pharmacology	8	Does
27	2	Reverse Pharmacology	8	Shows how
MODULE 28 — PHARMACEUTICAL QUALITY ASSURANCE — 19 OBJECTIVES, CEILING DOES				
28	1	Quality Assurance and Quality Management concepts	6	Shows how
28	2	Personnel Responsibilities	3	Shows how
28	3	Containers, Packing materials	4	Does
28	4	Good Laboratory Practices	3	Does
28	5	Document maintenance in pharmaceutical industry	3	Does
SEMESTER 5				
MODULE 5 — CRITICAL APPRAISAL OF UPARASA DRAVYA AND SADHARANA RASA DRAVYA — 24 OBJECTIVES, CEILING DOES				
5	1	Principle and Processing of Uparasa Dravya	12	Shows how
5	2	Principle and Processing of Sadharana Rasa	12	Does
MODULE 6 — CRITICAL APPRAISAL OF DHATU AND UPDHATU — 24 OBJECTIVES, CEILING DOES				
6	1	Principle and Processing of Dhatu / Metals	12	Does
6	2	Principle and Processing of Upadhatu	12	Does
MODULE 13 — KRUTANNA (AYURVEDIC FOOD PREPARATIONS) — 15 OBJECTIVES, CEILING DOES				
13	1	Classification of food substances and Formulations	3	Does
13	2	Customized food for treatment	6	Does
13	3	Ayurvedic Food Industry	3	Does

M	U	UNIT	OBJ	CEILING
13	4	Food processing technology traditional and modern methods	3	Shows how
MODULE 14 — AYURVEDIC FORMULATIONS FOR HOLISTIC WELLBEING, VITALITY AND BODY CARE — 11 OBJECTIVES, CEILING SHOWS HOW				
14	1	Rasayana-Personalized Rejuvenation Formulations	5	Shows how
14	2	Ayurvedic based cosmetic preparation	6	Shows how
MODULE 21 — CLINICAL APPLICATION OF RASOUSHADHI ACCORDING TO SROTAS AND IN SELECTED CHALLENGING CLINICAL CONDITIONS — 13 OBJECTIVES, CEILING DOES				
21	1	Clinical application of Rasoushadhis specific to Srotas	4	Shows how
21	2	Clinical application of Rasoushadhi in emergency conditions	4	Shows how
21	3	Rasoushadhi in specific, clinically challenging conditions	5	Does
MODULE 22 — YOGAS FROM DIFFERENT KALPANAS — 27 OBJECTIVES, CEILING DOES				
22	1	SOP of Vati, Gutika, Guggulu and Varti Kalpas and their therapeutic applications	6	Shows how
22	2	SOP and Clinical applications of Avaleha and Khanda kalpana	5	Does
22	3	SOP of Sneha Kalpana and its clinical significance	5	Shows how
22	4	SOP and clinical applications of Arka and Dravakam in various disorders	5	Does
22	5	SOP and therapeutic applications of Sandhana Kalpana	6	Does
MODULE 29 — PHARMACY PRACTICE — 18 OBJECTIVES, CEILING DOES				
29	1	Hospital Pharmacy and Community Pharmacy	7	Does
29	2	Therapeutic Drug Monitoring	7	Shows how
29	3	Pharmacy and therapeutic committee	4	Does
MODULE 30 — PHARMACEUTICAL MICROBIOLOGY — 18 OBJECTIVES, CEILING DOES				
30	1	Introduction to microbes with applications of sterilization	5	Shows how
30	2	Sources of contamination	6	Does
30	3	Types of spoilage and use of antimicrobial agents	7	Does
SEMESTER 6				
MODULE 7 — SUDHA VARGA, SIKATA VARGA, VISHA VARGA DRAVYAS — 22 OBJECTIVES, CEILING DOES				
7	1	Characterization, Pharmaceutical Processing, and Clinical Applications of Sudha Varga Dravya	7	Shows how
7	2	Characterization, pharmaceutical processing, and clinical applications of Sikata Varga Dravya	7	Shows how
7	3	Pharmaceutical processing, phytopharmacology and clinical applications of Visha Varga dravya	8	Does
MODULE 8 — APPRAISAL OF RATNA AND UPARATNA AND THERAPEUTIC SIGNIFICANCE OF PISHTI AND DRUTI KALPANA — 13 OBJECTIVES, CEILING DOES				
8	1	Mineralogy, Crystal chemistry,and Pharmacotherapeutics of Ratna-Uparatna Varga	7	Does
8	2	SOP and therapeutic applications of Pishti, Gandhaka Druti and Gandhaka Taila kalpana	6	Shows how
MODULE 15 — DISEASE CONDITION SPECIFIC SELECTION OF FORMULATION — 12 OBJECTIVES, CEILING SHOWS HOW				
15	1	Application of Yukti Pramana in Formulation Selection	2	Knows how
15	2	Selection of formulations for Speciality practice in Ayurveda	3	Shows how

M	U	UNIT	OBJ	CEILING
15	3	Routes of Drug Administration	2	Shows how
15	4	Designing formulations for Shodhana and Shamana Chikitsa	2	Shows how
15	5	Innovation in Ayurvedic Formulations	3	Shows how
MODULE 16 — CURRENT TRENDS IN AYURVEDIC PHARMACEUTICAL INDUSTRY. — 12 OBJECTIVES, CEILING DOES				
16	1	Key Competencies Required for Employment in Pharmacy	6	Does
16	2	Internship/ Visits to GMP certified Pharmacies	3	Shows how
16	3	Quality Assurance and Quality Control (QAQC) Laboratory Training	3	Shows how
MODULE 23 — SOP AND CLINICAL APPLICATIONS OF CHURNA KALPANA(POWDER DOSAGE FORMS) — 26 OBJECTIVES, CEILING DOES				
23	1	SOP and clinical utility of Churna Kalpana	4	Does
23	2	Standardization, analytical profiling, and clinical application of Kshara	5	Does
23	3	Lavana Kalpana with special reference to pharmaceutico-analytical aspects and therapeutic attributes	5	Does
23	4	Satva Kalpana SOP and clinical significance	5	Shows how
23	5	Therapeutic attributes of Masi Kalpana	7	Does
MODULE 24 — THERAPEUTIC SIGNIFICANCE OF TOPICAL DOSAGE FORMS — 11 OBJECTIVES, CEILING SHOWS HOW				
24	1	Ophthalmic, Otic, and Buccal dosage forms and their clinical applications1.SOP for Ophthalmic, Otic and buccal dosage forms	4	Shows how
24	2	Standardisation of Lepa and Malahara preparations and its application in various disorders	3	Shows how
24	3	Standard Operating Procedure (SOP) for Upanaha, Dhupa, Dhumapana, and Avachurnana Kalpa and their therapeutic applications	4	Shows how
MODULE 31 — REGULATORY ACTS AND RULES APPLICABLE TO AYURVEDIC DRUGS — 31 OBJECTIVES, CEILING DOES				
31	1	Chapter I of the Drugs and Cosmetics Act, 1940 and Rules 1945	3	Does
31	2	Chapter II - Regulatory Bodies	3	Does
31	3	Chapters III, IV, and IV-A - Regulation of Drugs and Cosmetics	8	Does
31	4	Schedules to Drugs and Cosmetics Rules related to ASU drugs and cosmetics	5	Does
31	5	Pharmaceutical observation of drugs and cosmetics ASU drugs	3	Does
31	6	Miscellaneous Regulatory Acts related to drugs and cosmetics	9	Does
MODULE 32 — ENTREPRENEURSHIP AND START-UPS — 14 OBJECTIVES, CEILING DOES				
32	1	Entrepreneurship and Startup	3	Does
32	2	Process of entrepreneurship development	3	Does
32	3	Project Proposal for new Invention/startup	2	Does
32	4	Intellectual Property Rights and Patents	6	Does

Module ceilings are shown in the coloured band; unit ceilings in the rows beneath. A unit may sit below its module's ceiling, and most do.

PART C · APPENDIX C3

The instruments this curriculum names, by tier

Every assessment abbreviation that appears in the curriculum, sorted by the highest tier it can honestly certify.

HIGHEST TIER IT REACHES	INSTRUMENTS THE CURRICULUM NAMES	WHAT IT CAN AND CANNOT SHOW
Knows	MCQ, Spotting, T-OBT	Recall only. Cannot show reasoning or performance.
Knows how	LAQ, SAQ, Structured viva, ABQ, Case study	Reasoning on paper or aloud. Cannot show that the student can perform.
Shows how	OSPE, Practical examination, Checklist	Performance when watched. The lowest instrument that settles a doing objective.
Does	Log book, Record book, Portfolio, Presentation	Independent practice over time. The only evidence that reaches the top tier.

Counts are how many objectives name each instrument. An instrument may always be used below its tier; it may never be used above it.

PART C · APPENDIX C4

Sources, and a note on how this book was made

The primary sources for the framework, the curriculum this book is built on, and a note on what was and was not altered.

THE FRAMEWORK

- Miller, G. E., 'The assessment of clinical skills / competence / performance', *Academic Medicine*, 65(9), 1990. The original four-tier paper.
- Cruess, R. L., Cruess, S. R. and Steinert, Y., 'Amending Miller's Pyramid to Include Professional Identity Formation', *Academic Medicine*, 91(2), 2016. The proposed fifth tier, *Is*.
- Biggs, J., 'Enhancing teaching through constructive alignment', *Higher Education*, 32, 1996. The source of the alignment principle in Chapter 3.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words. Plain-English spelling errors have been corrected and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered, and no subject content has been added, removed or reworded.

National Commission for Indian System of Medicine, Board of Ayurveda. *Curriculum for MD/MS Ayurveda, Semesters III to VI — Rasashastra and Bhaishajya Kalpana (AYPG-RB).* New Delhi: NCISM. Tables 2, 3, 4, 5 and 6 are the source of every objective, tier, hour and mark in this book. All 577 objectives and 304 activities reconcile against the curriculum's own credit framework: 1792 teaching hours plus 128 of modular assessment give the 1920 notional hours of 64 credits, and each semester independently totals 480.

The counts and tables in Part C are generated from the same parsed data as Part B, so the two cannot disagree.